Fundamentals of Ultra-Low Voltage Embedded Memory Design

Eric Karl

Intel Fellow, Director of Embedded Memory Circuits & Technology
Technology Development, Intel Corporation

Live Q&A Session: February 20th, 2022, 7-7:20 AM PST

Introduction

Eric Karl, Intel Fellow, Technology Development

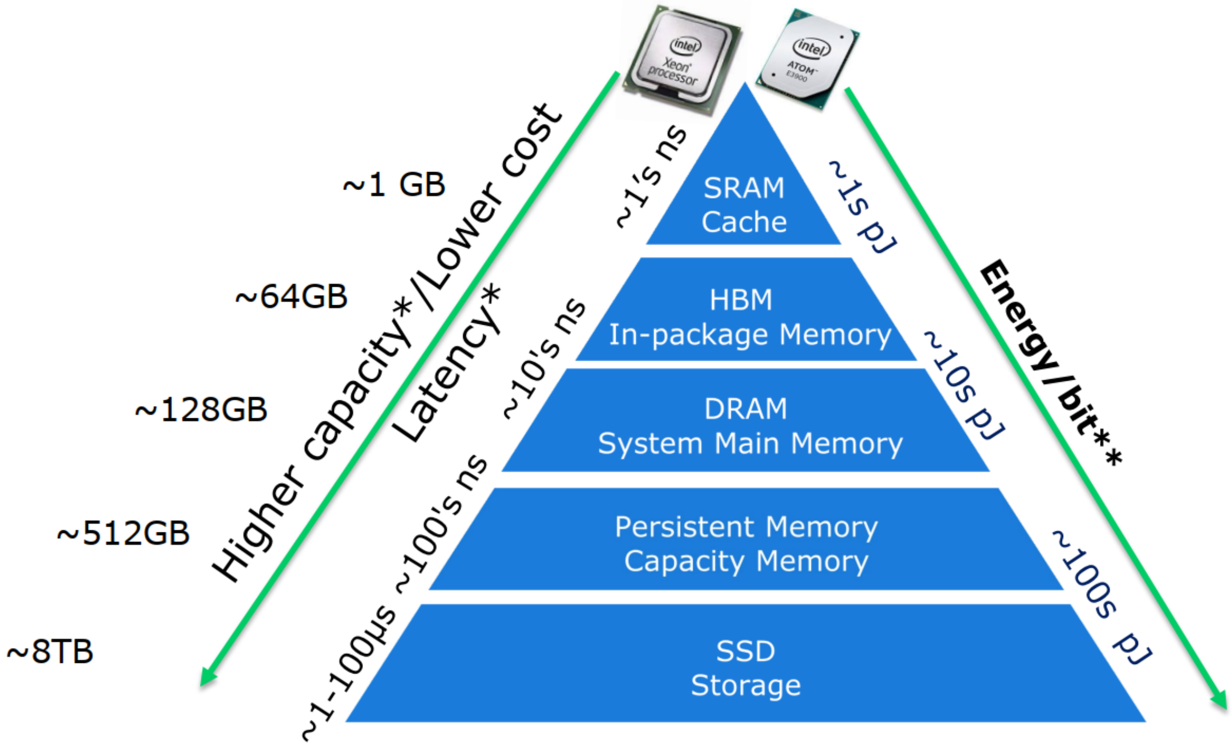
- BS & MS in Electrical Engineering from Univ. of Michigan
- Ph.D. in Electrical Engineering from Univ. of Michigan
- SRAM Design Lead at Intel TD, 2008-2013
- Memory Circuit Technology Leader at Intel TD, 2013-2020
- Director of Advanced Design at Intel TD, 2020-Present

- Area of Expertise: Memory Design & DTCO
  - Embedded Memory Design – SRAM, Register Files, Fuse/OTP
  - Advanced Logic Process Technologist
  - Design Technology Co-optimization
  - Digital Logic Library Development & Optimization

Outline

- Embedded Memory Trends
- SRAM: The Embedded Memory Workhorse
- Register Files: Special, High-Performance Memory for "XPU"
- Logic Sequentials: Ultra-Low Voltage Flip-Flops
- System Design: Putting it all together
- Conclusion

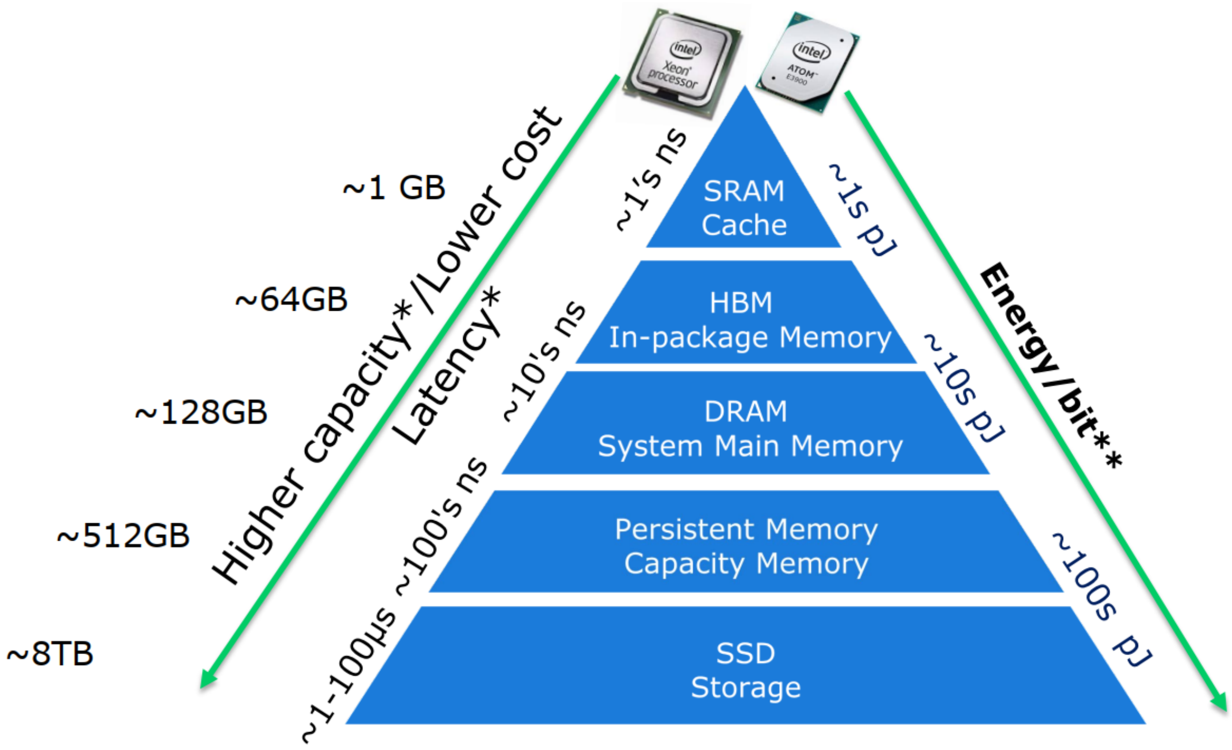
Modern Memory Hierarchy



In a hierarchical memory system, the entire addressable memory space is available in the largest, slowest memory and incrementally smaller and faster memories, each containing a subset of the memory below it, proceed in steps up toward the processor.

Source: Shanthi, Web

Modern Memory Hierarchy



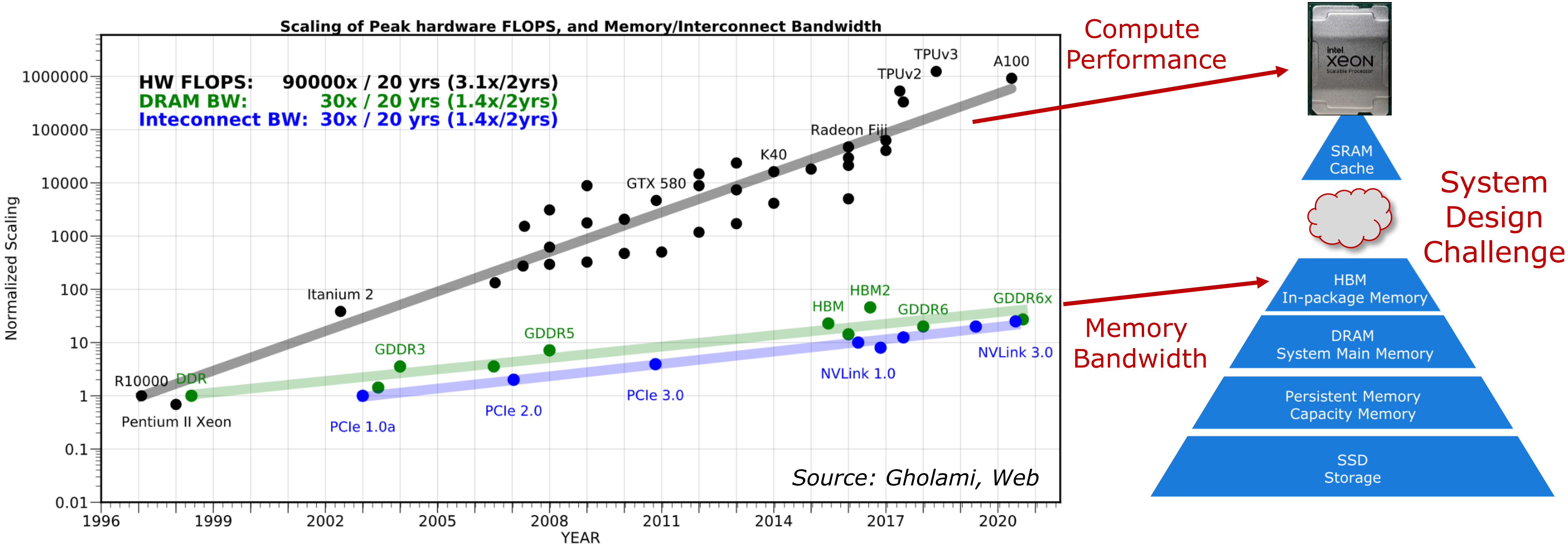
This hierarchical organization of memory works because of the ***Principle of Locality.*** Programs access a relatively small portion of the address space at any moment. There are two different types of locality:

- ***Temporal Locality***: If an item is referenced, it will tend to be referenced again soon
- ***Spatial Locality***: If an item is referenced, items whose addresses are close by tend to be referenced soon

Source: Shanthi, Web

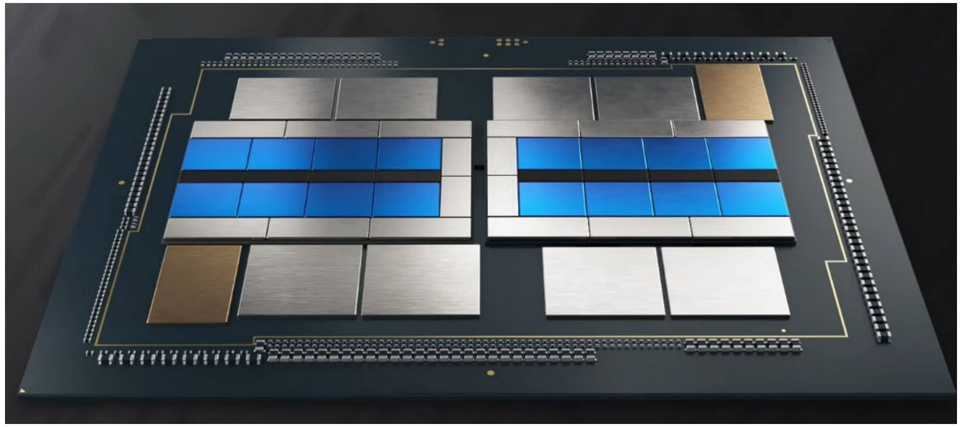
- Modern hardware relies upon temporal and spatial locality of memory accesses to create a high-capacity, high performance and energy efficient virtual memory system

System Memory Bandwidth Challenges

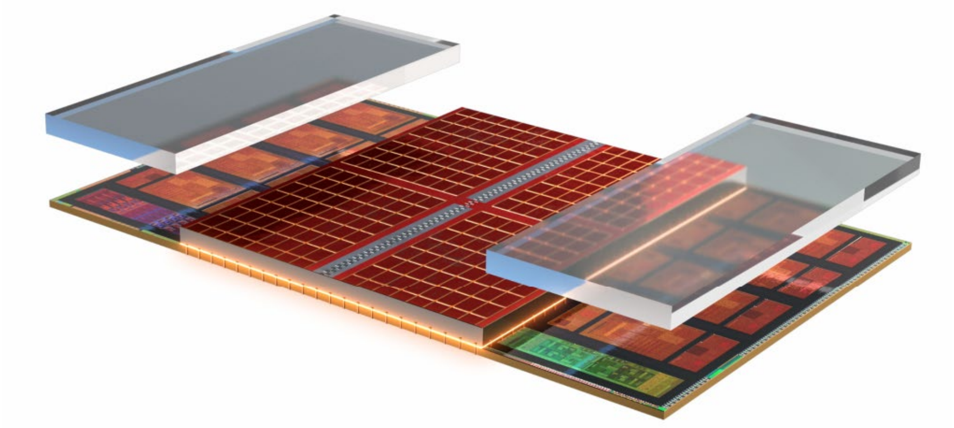


- Peak compute performance has improved at a much faster rate than bandwidth to high-capacity system memory over the last decade

Bandwidth Gap driving SRAM Cache



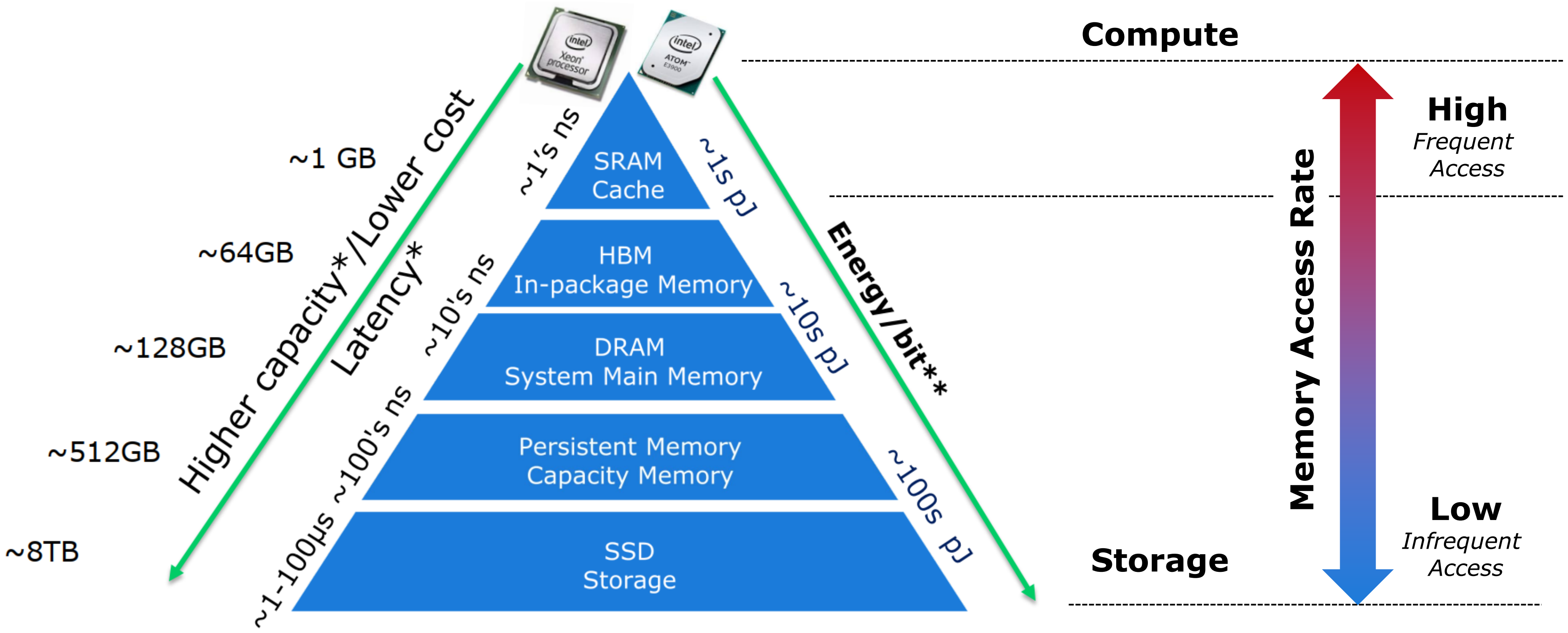
Intel Ponte Vecchio
16 Compute Tiles
64MB Total L1 Cache
408MB Total L2 Cache
8 HMB2e Tiles



AMD Zen3 with 3D V-Cache
8 Compute Cores
512kB Total L1 Cache
4MB Total L2 Cache
Up to 96MB Shared L3 Cache

- Bandwidth gap to main memory is driving **increased SRAM in package**
- Latest products exceeding Gbit levels of integration in last level SRAM cache

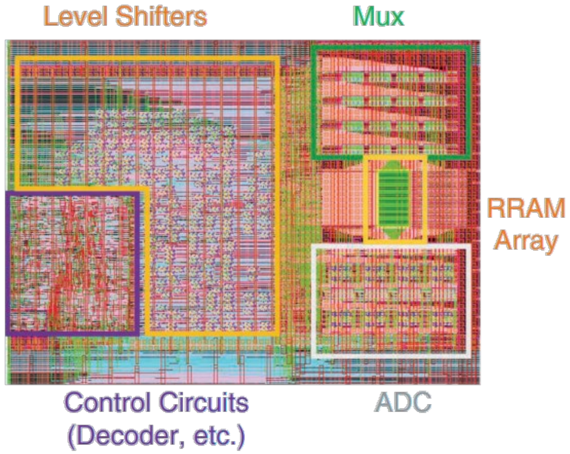
Demand for Energy Efficient Cache



- Compute and associated high activity embedded cache faces ever-increasing pressure deliver **sustained energy efficiency improvement and operate at ultra-low voltages**

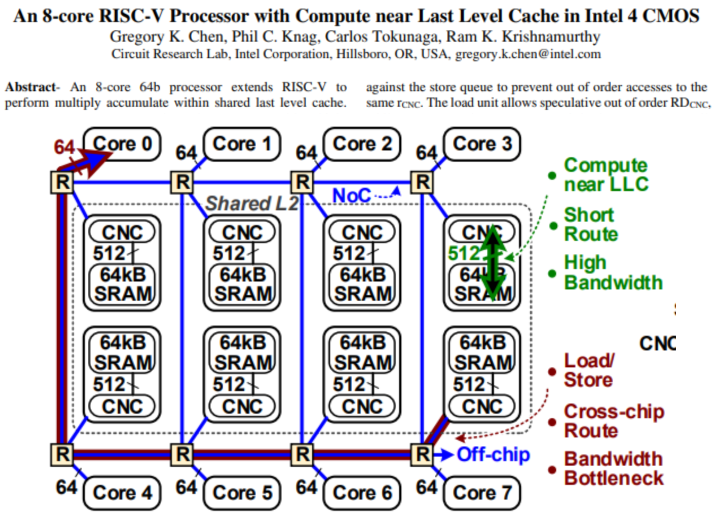
Memory-Centric Computing

Compute inside Memory



- Reduced Data Movement Energy
- Energy Efficient Compute using Array Structure

Compute near SRAM Cache



An 8-core RISC-V Processor with Compute near Last Level Cache in Intel 4 CMOS

Gregory K. Chen, Phil C. Knag, Carlos Tokunaga, Ram K. Krishnamurthy

Circuit Research Lab, Intel Corporation, Hillsboro, OR, USA, gregory.k.chen@intel.com

Abstract- An 8-core 64b processor extends RISC-V to perform multiply accumulate within shared last level cache. against the store queue to prevent out of order accesses to the same r_{CNC}. The load unit allows speculative out of order RD_{CNC},

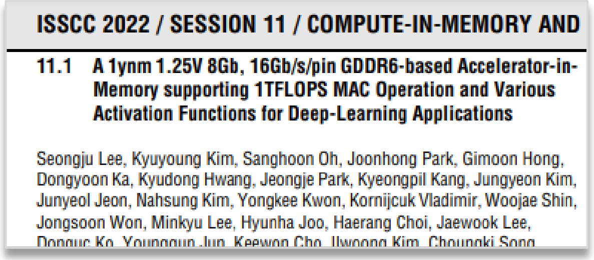
- Reduced Data Movement Energy

Compute near DRAM Memory

Aquabolt-XL: Samsung HBM2-PIM with in-memory processing for ML accelerators and beyond

Jin Hyun Kim, Shin-haeng Kang, Sukhan Lee, Hyeonsu Kim, Woongjae Song, Yuhwan Ro, Seungwon Lee, David Wang, Hyunsung Shin, Bengseng Phuah, Jihyun Choi, Jinin So, YeonGon Cho, JoonHo Song, Jangseok Choi, Jeonghyeon Cho, Kyomin Sohn, Youngsoo Sohn, Kwangil Park, and Nam Sung Kim

Samsung Electronics



ISSCC 2022 / SESSION 11 / COMPUTE-IN-MEMORY AND

11.1 A 1ynm 1.25V 8Gb, 16Gb/s/pin GDDR6-based Accelerator-in-Memory supporting 1TFLOPS MAC Operation and Various Activation Functions for Deep-Learning Applications

Seongju Lee, Kyuyoung Kim, Sanghoon Oh, Joonhong Park, Gimoon Hong, Dongyoon Ka, Kyudong Hwang, Jeongje Park, Kyeongpil Kang, Jungyeon Kim, Junyeol Jeon, Nahsung Kim, Yongkee Kwon, Kornijcuk Vladimir, Woojae Shin, Jongsoon Won, Minkyu Lee, Hyunha Joo, Haerang Choi, Jaewook Lee,

- Reduced Data Movement Energy

- ☐ Emerging Memory-Centric computing models **targeting energy efficiency**
- ☐ Most schemes will require basic memory operation **at ultra-low voltages**

Outline

- Embedded Memory Trends
- SRAM: The Embedded Memory Workhorse
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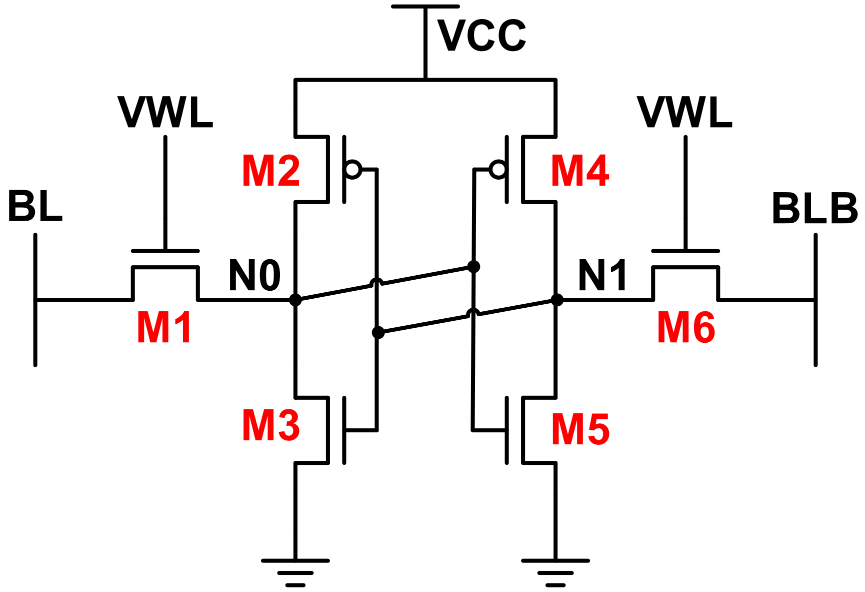
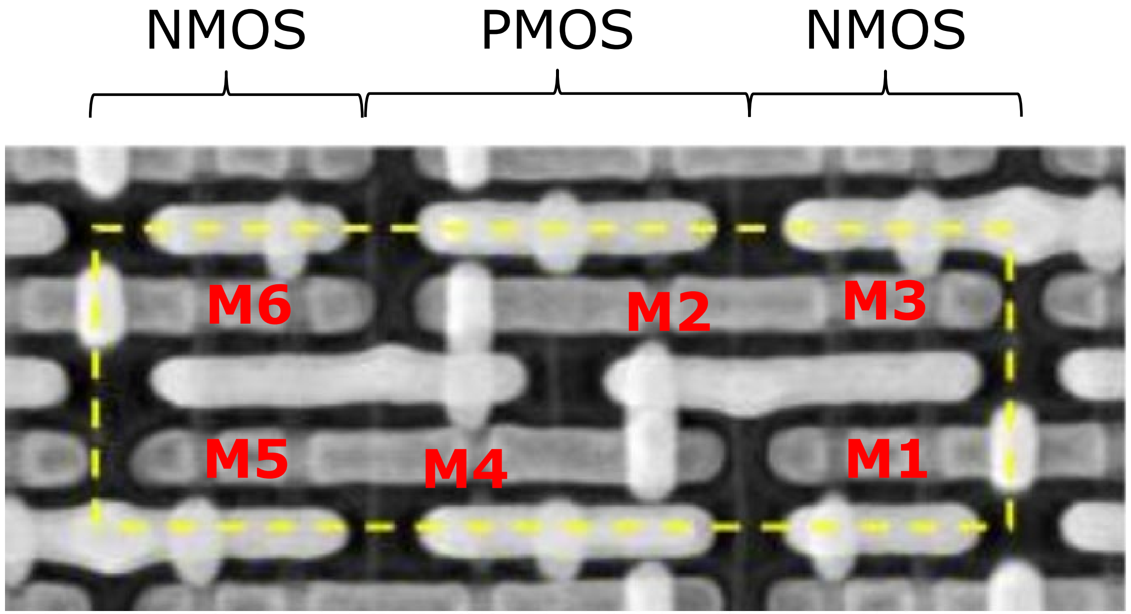
Static Random Access Memory (SRAM)

- The workhorse embedded memory element in advanced CMOS technology
- Bi-stable, volatile storage element
- Can be constructed without logic process cost adders in advanced nodes

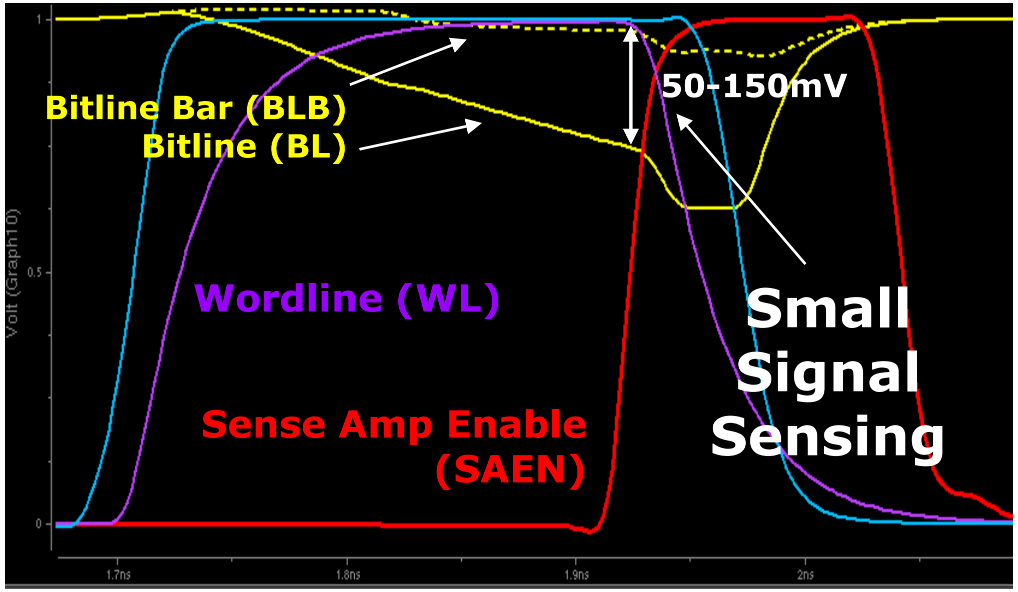
M1 / M6: Pass-gate (PG)
M2 / M4: Pull-up (PU)
M3 / M5: Pull-down (PD)

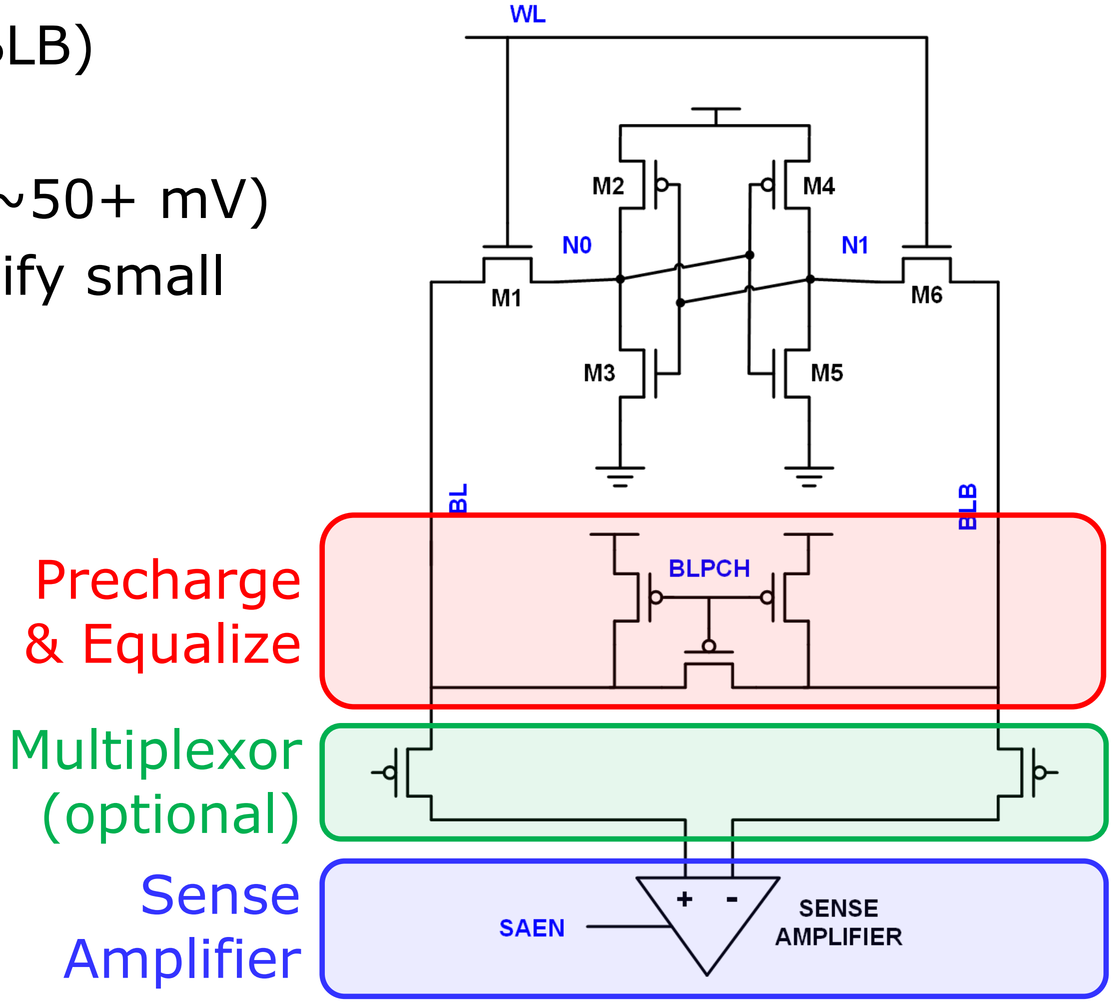




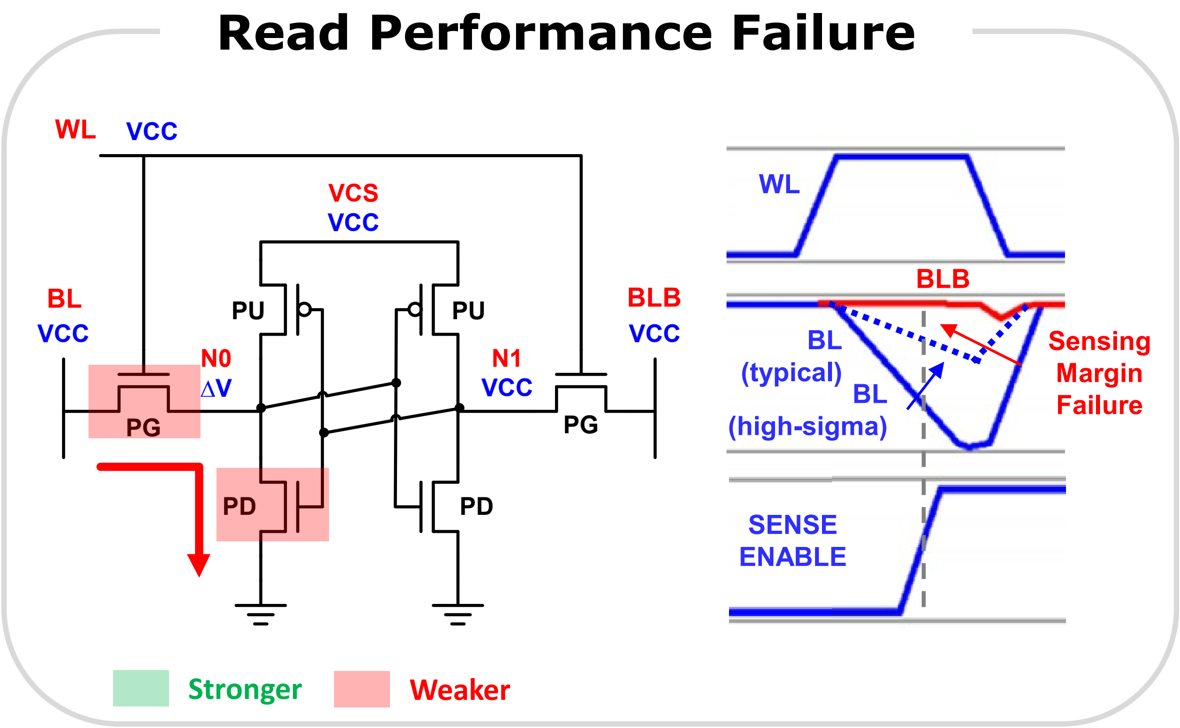
6T SRAM Read: Basic Operation

- Precharge and equalize bitlines (BL & BLB)
- Assert bitcell wordline signal (WL)
- Develop differential signal on bitlines (~50+ mV)
- Trigger sense amplifier (SAEN) to amplify small signal to large output signal





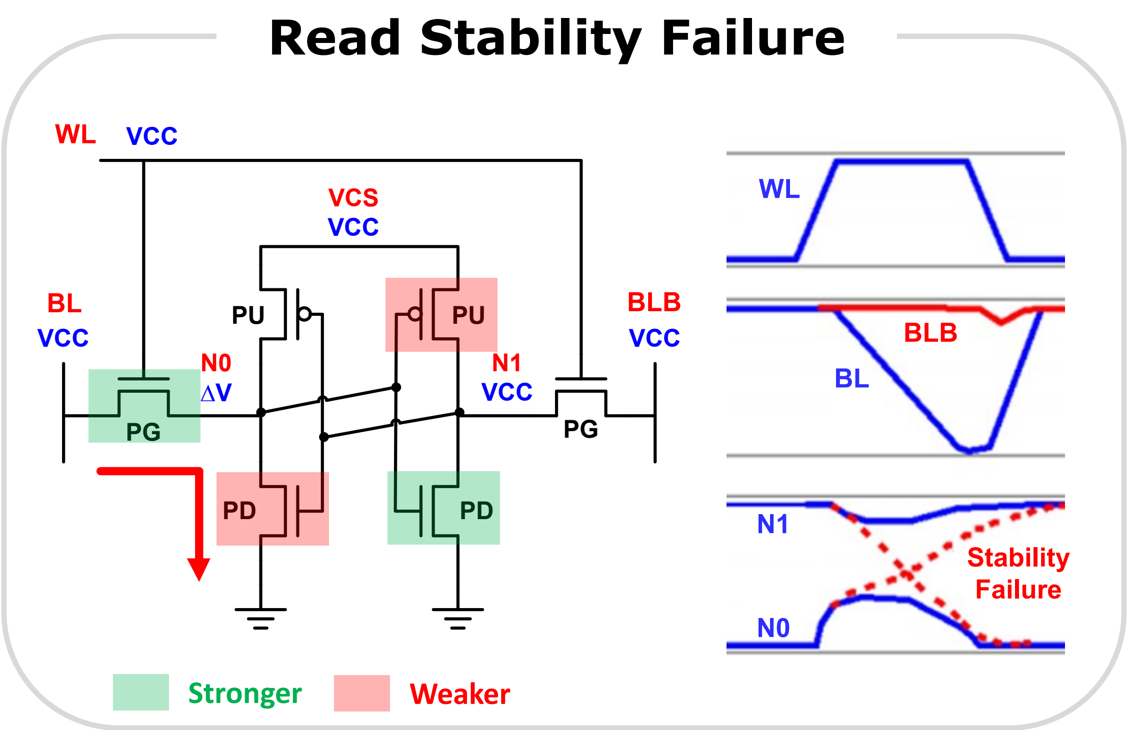
6T SRAM Read: Performance Concerns

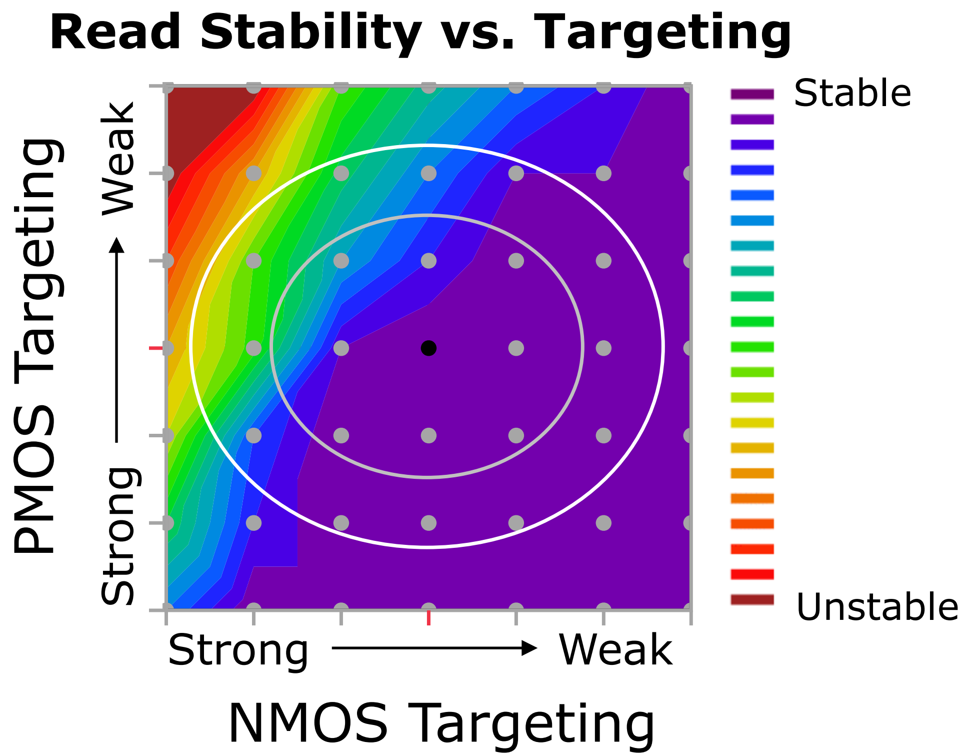


- Read Performance Failures are influenced by variation in the following parameters:
  - Bitcell Read Current
  - Bitline Capacitance
  - Bitline Resistance
  - Wordline RC
  - Sense Amplifier Mismatch

- Read performance failures in SRAM arrays are driven by insufficient voltage differential development between BL and BLB nodes leading to sense amplifier failures

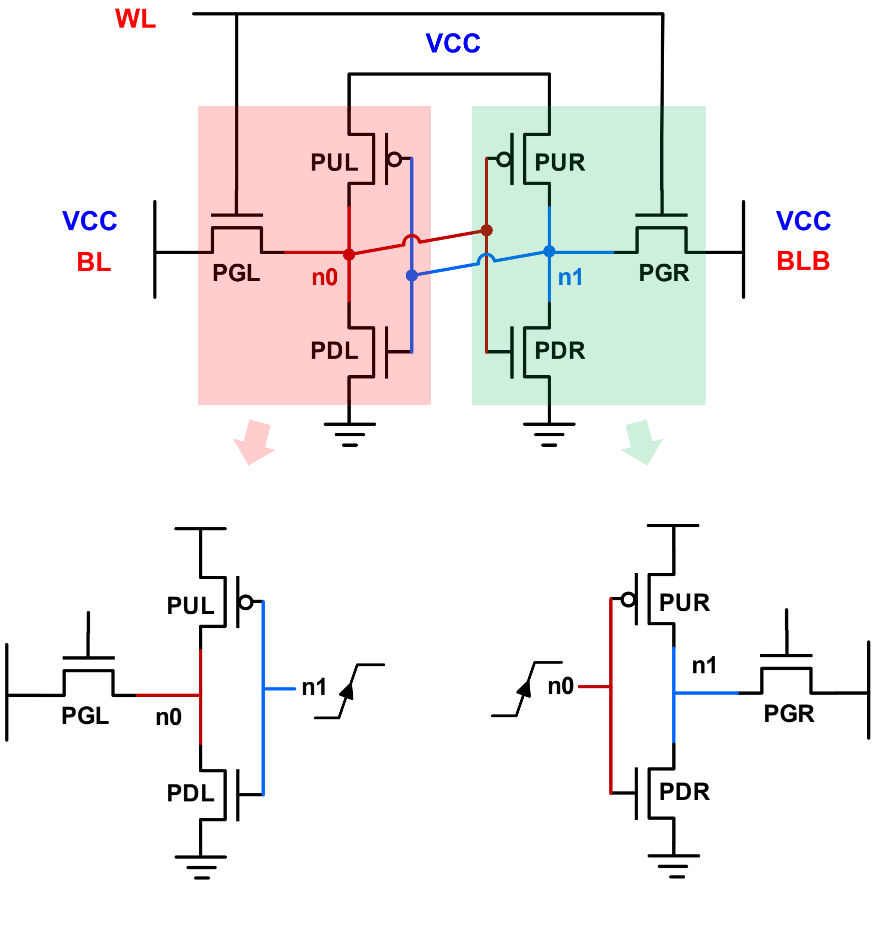
6T SRAM Read: Stability Concerns





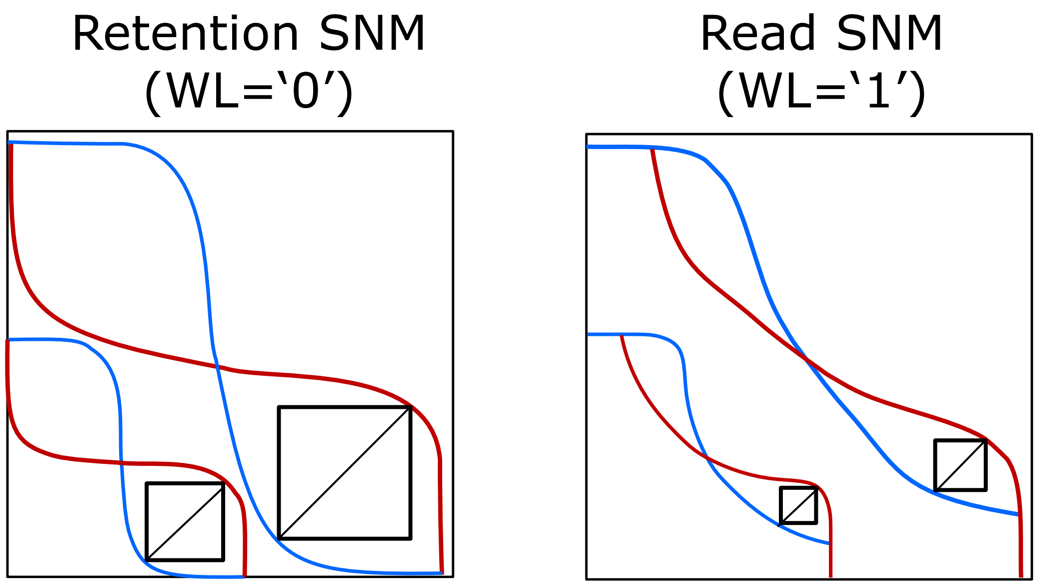
- Read stability failures result in a bitcell that loses state during a read operation due to charge injection from bitline to internal nodes

Static Noise Margin Analysis (SNM)



Source: Seevinck, JSSC 1987

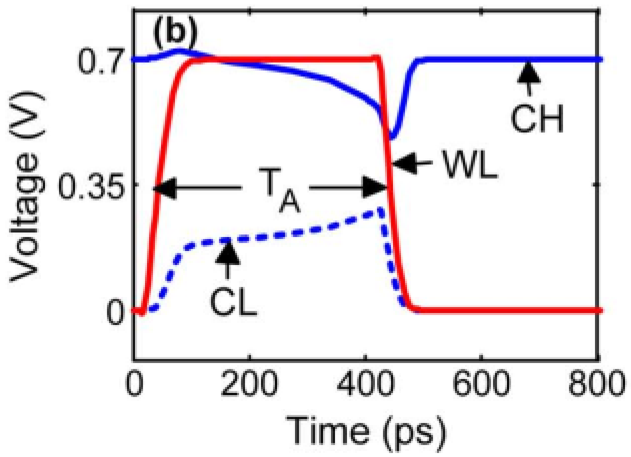
- Left and Right side are decoupled and BL/BLB nodes are held at VCC
- Voltage transfer curves are superimposed to find SNM (the largest square that fits between the transfer curves)



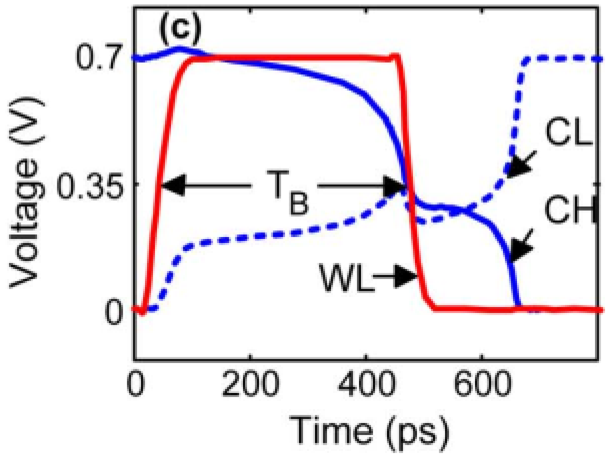
Dynamic Stability

- Static noise magin (SNM) analysis is a safe worst-case on read stability
  - Static bitline voltage can be pessimistic if array design allows discharge
  - Read stability “state flips” have a latency → static assumption is worst case
- Dynamic stability margin analysis is more accurate for modeling to absolute lowest voltages possible

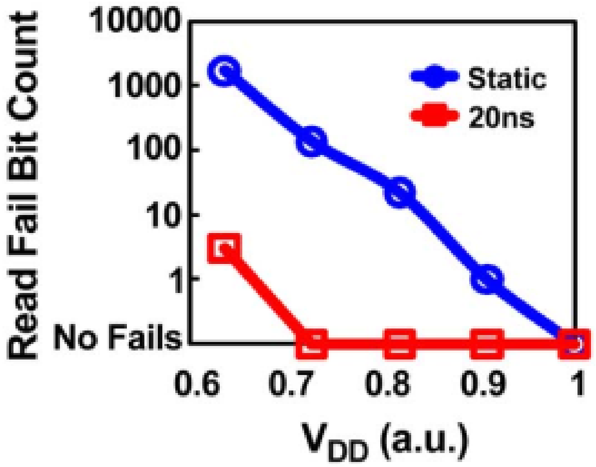
Read Stable



Read Unstable

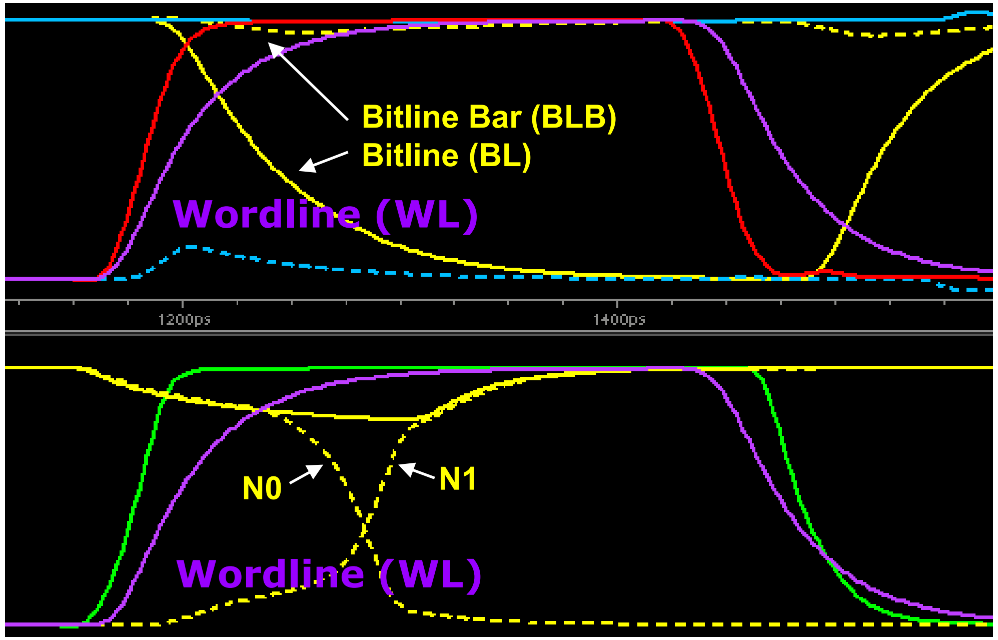


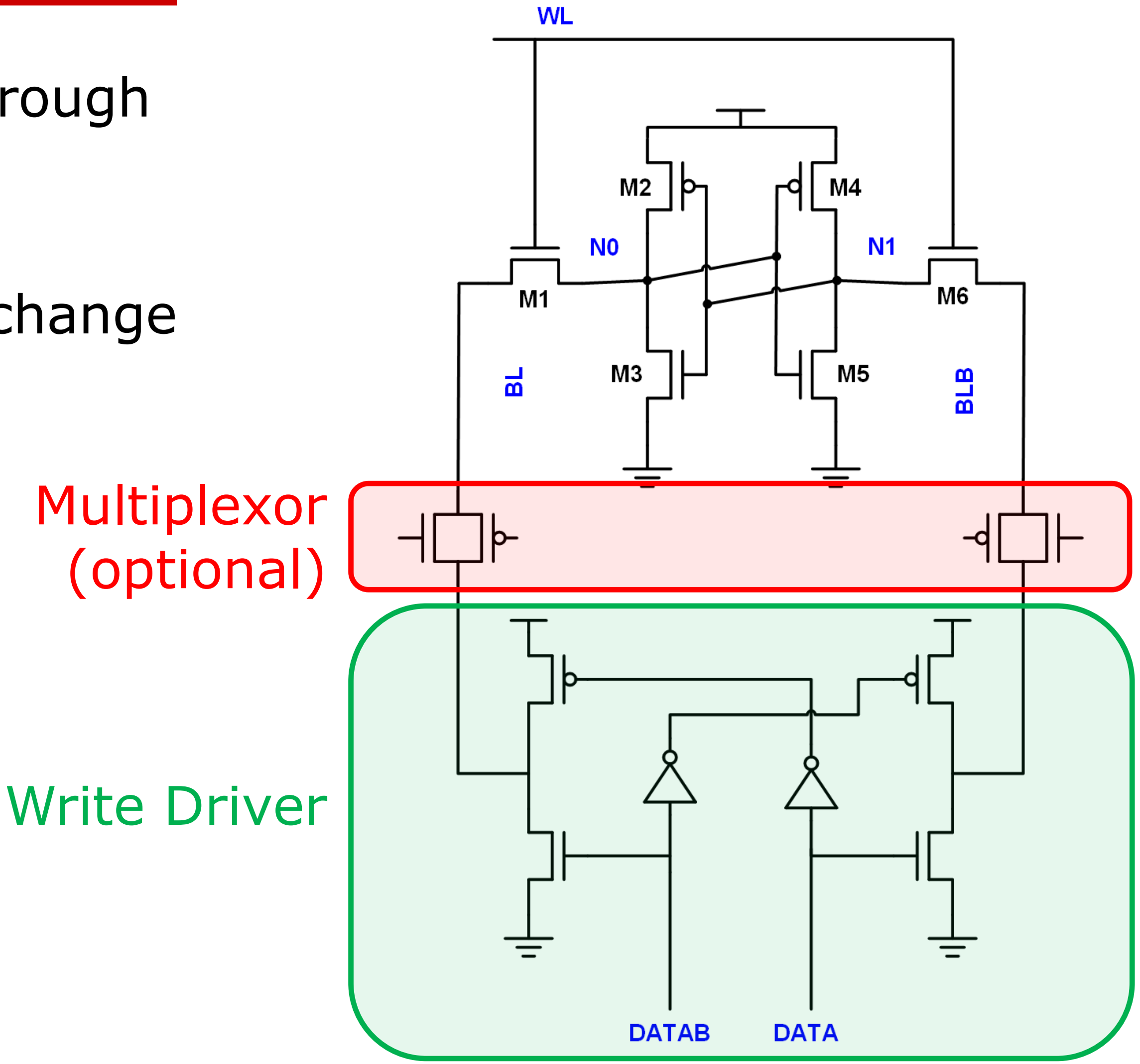
Measured 45nm SRAM



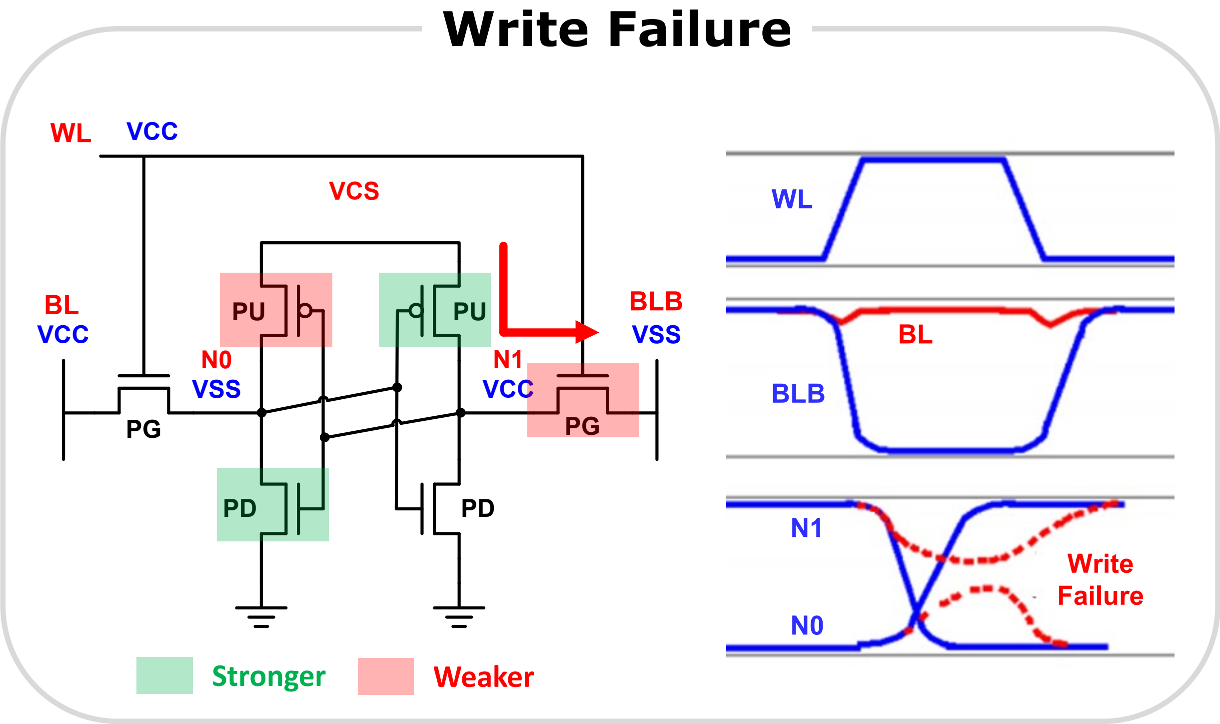
6T SRAM Write: Basic Operation

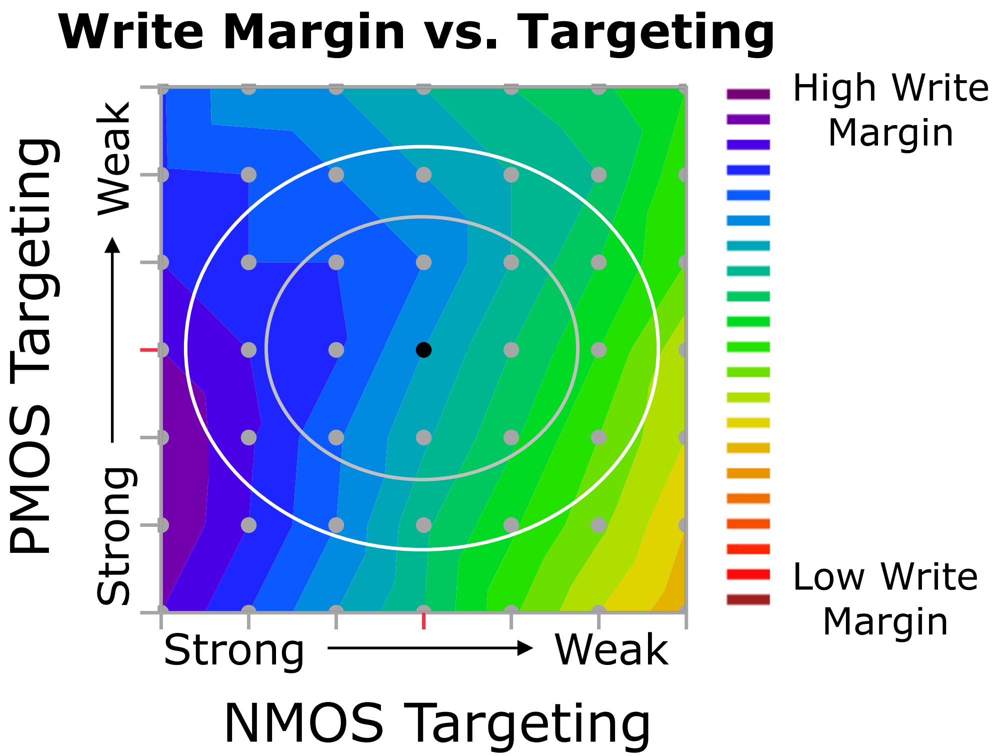
- Drive bitline pair to opposite states through write driver and multiplexor circuit
- Assert bitcell wordline control signal
- Internal nodes of bitcell (N0 and N1) change state to reflect new written data





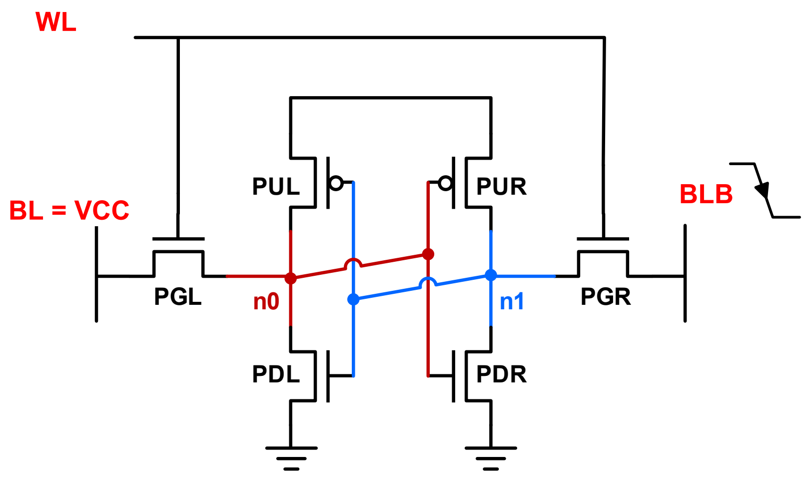
6T SRAM Write: Margin Concerns

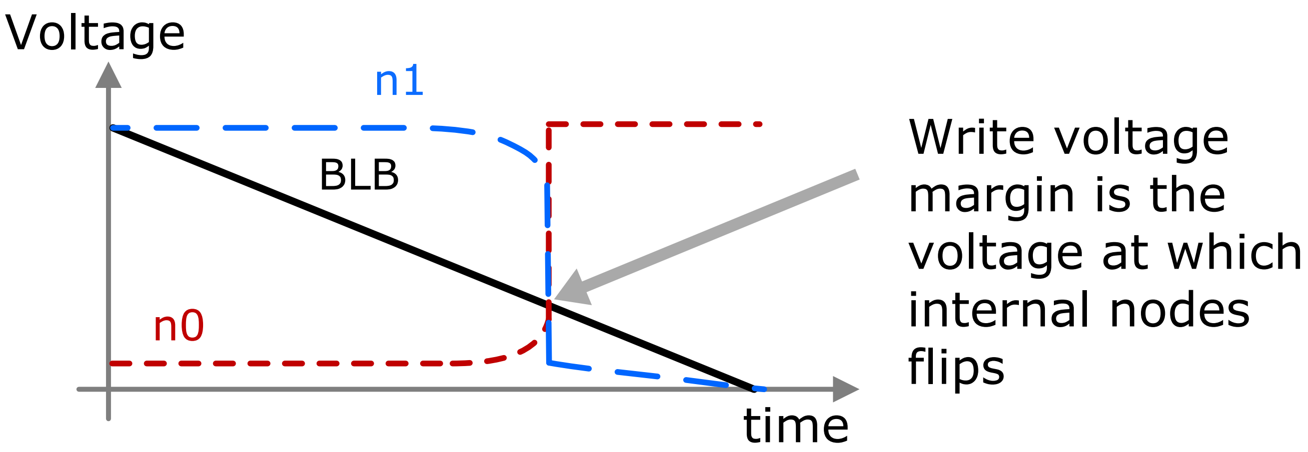




- Write margin failures occur when NMOS passgate devices are incapable of overwriting state held by the cross-coupled inverter pair
- Driven by NMOS passgate contention with PMOS pullup device

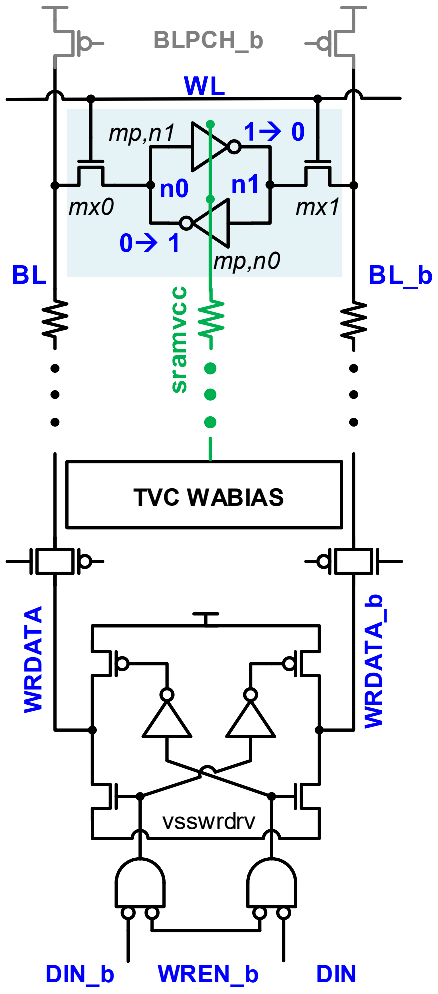
Static Write Voltage Margin (WVM)



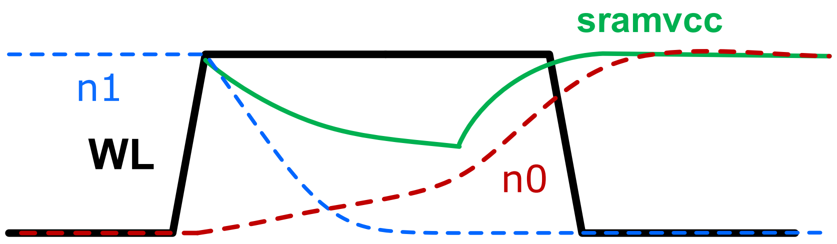


- In static write voltage margin measurement, bitline (BL) is held at VCC and bitline bar (BLB) is ramped down
- Random Variation analysis is usually applied to assess functionality under specific process skew, voltage and temperature conditions
- Write Voltage Margin is the bitline voltage at which the bit flips

Dynamic Write Margin



- Static write voltage margin is **usually optimistic**
- Dynamic write margin simulation is able to best capture the interaction between the bitcell and peripheral circuits
  - Time-dependent writability (i.e. write-at-speed)
  - Bitline resistance and capacitance impact
  - Peripheral circuit impact
  - Cross-coupling effect from neighboring signals

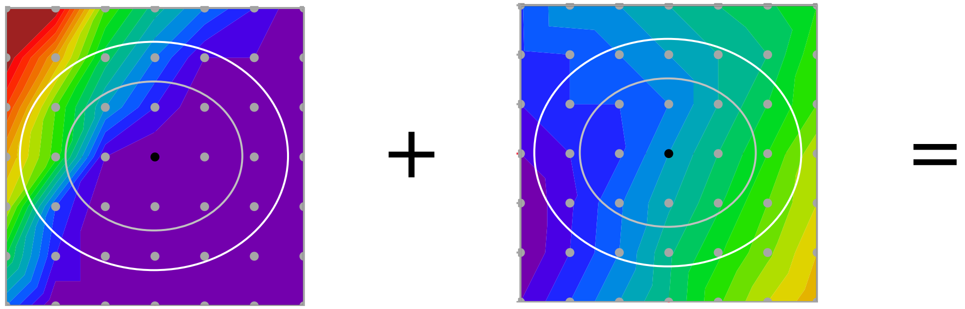


Simulation waveform of a write operation with TVC is shown

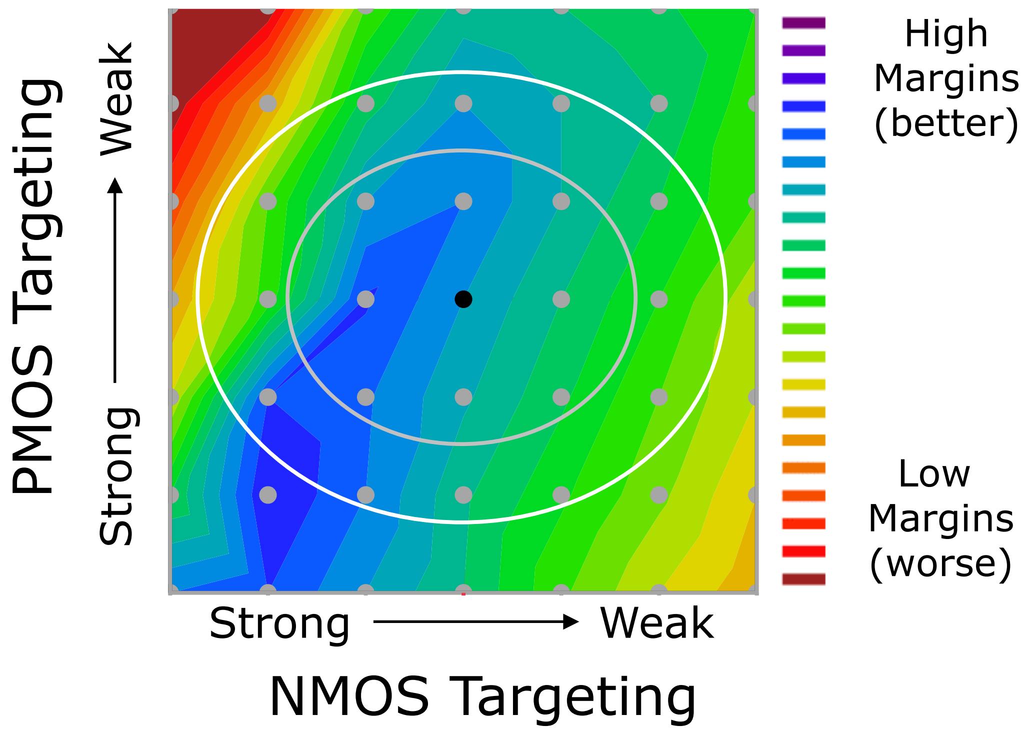
6T SRAM: Bitcell Design Considerations

SRAM design must balance *conflicting requirements* for stability and write margin

Stability Margin **Write Margin**

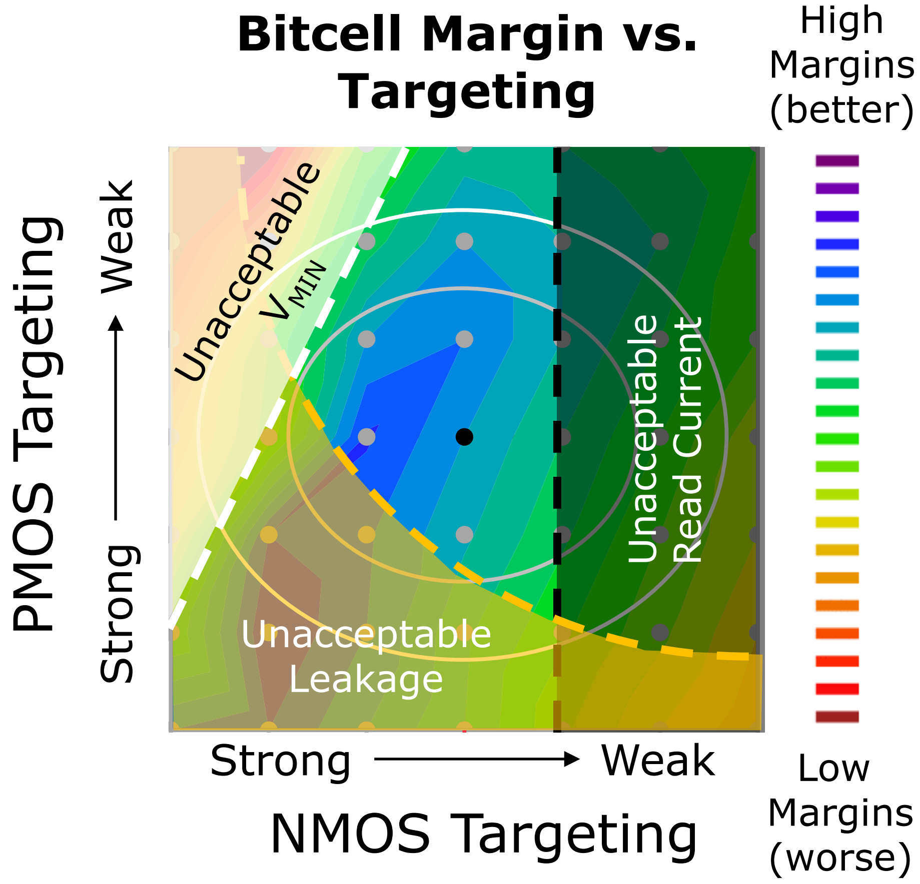


Bitcell Margin vs. Targeting



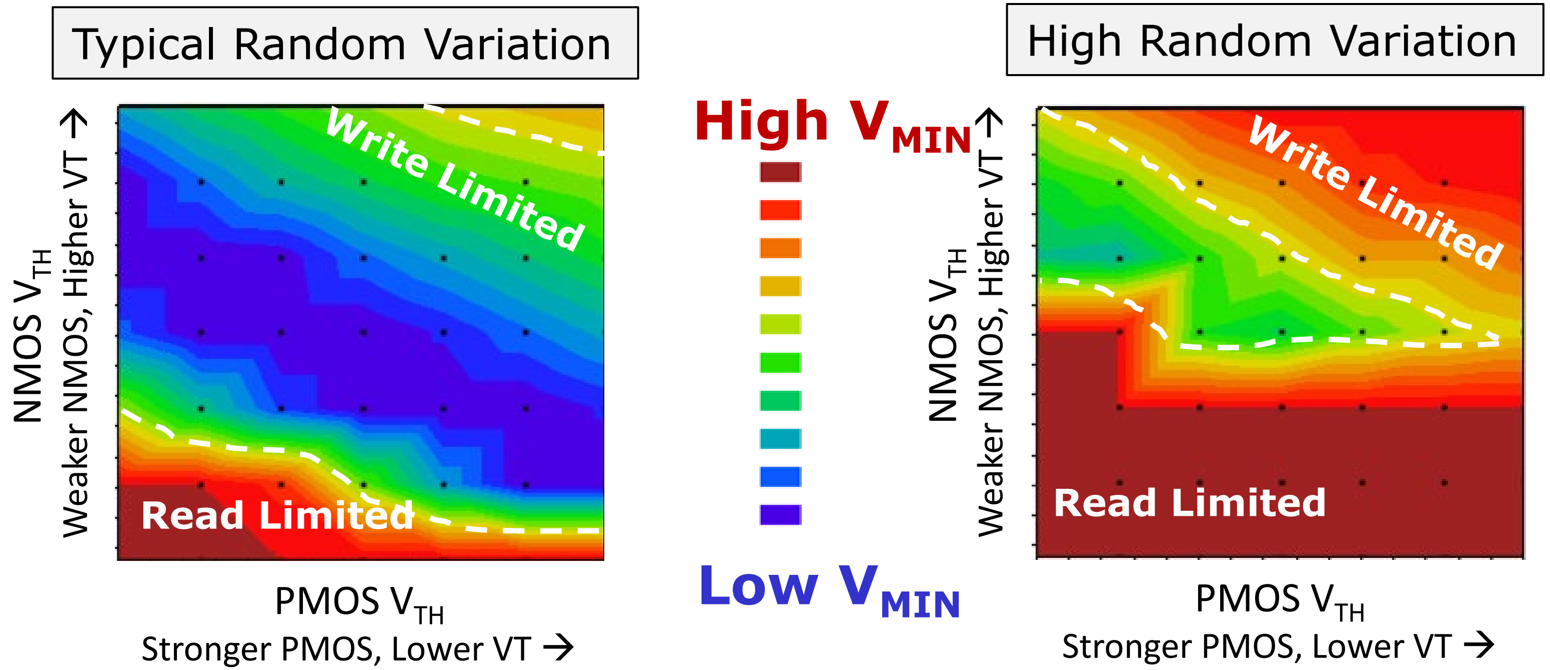
- Device sizing and targeting in SRAM is central to delivering adequate margins
- Higher margin cells can operate at lower voltages without electrical failures

6T SRAM: Bitcell Design Considerations



- Bitcell read current and leakage requirements place additional constraints on cell design
  - NMOS targeting is constrained by minimum acceptable read current (cell performance)
  - NMOS and PMOS targeting is constrained on the strong side by maximum acceptable leakage current
- Landing zone for SRAM to meet V_{MIN} (margins-driven), performance and leakage requirements is challenging in advanced CMOS technologies

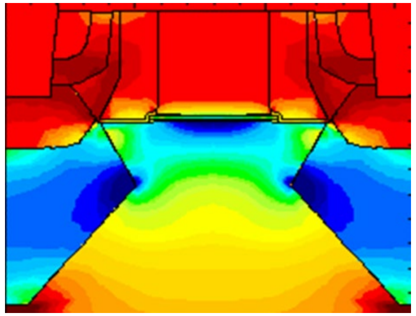
Variation: Random vs. Systematic



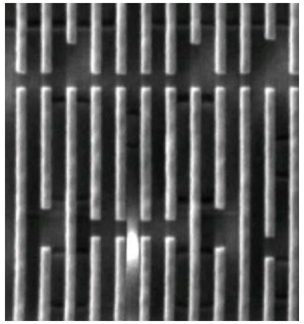
- Random variation determines the contours of constant memory V_{MIN}
  - Random variation is a top concern for memory V_{MIN} in scaled technology
- Memory V_{MIN} target and random variation dictates the process systematic variation landing zone

Systematic Variation Challenges

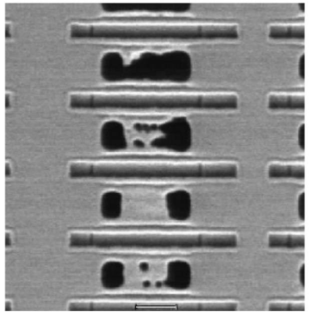
Stress



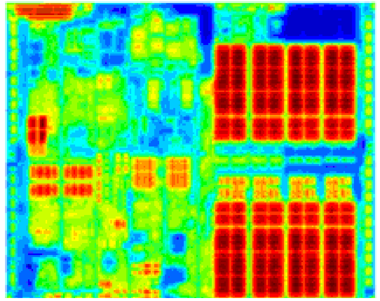
Litho



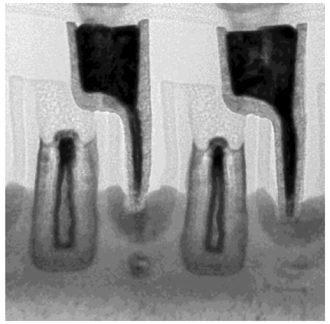
Polish

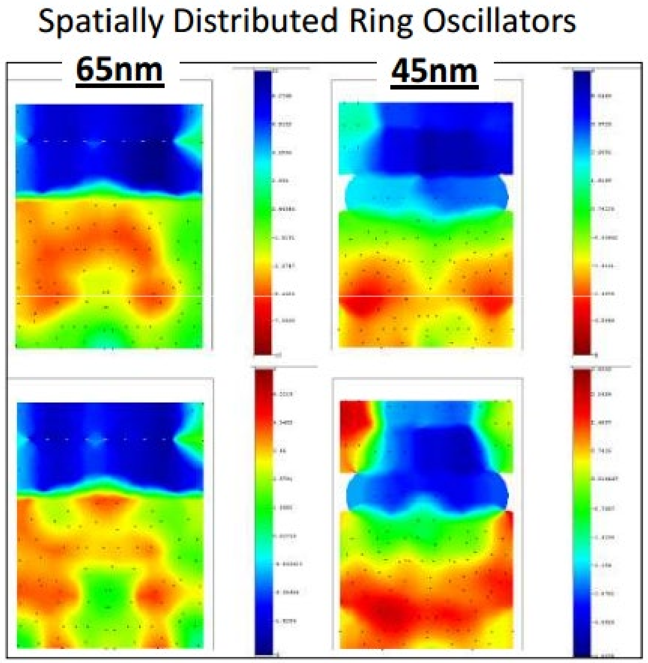


Anneal



Etch



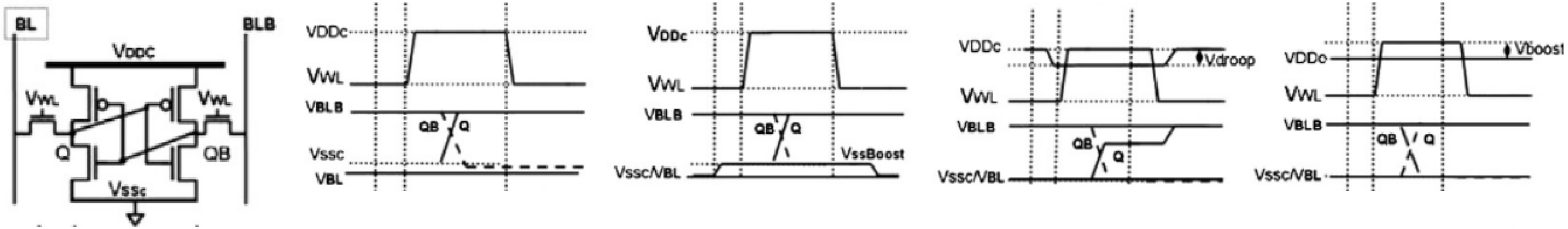


- New technologies continue to introduce new forms of systematic variation
- Challenging to meet goals of memories accounting for systematic variation

Write Assist Circuits for SRAM

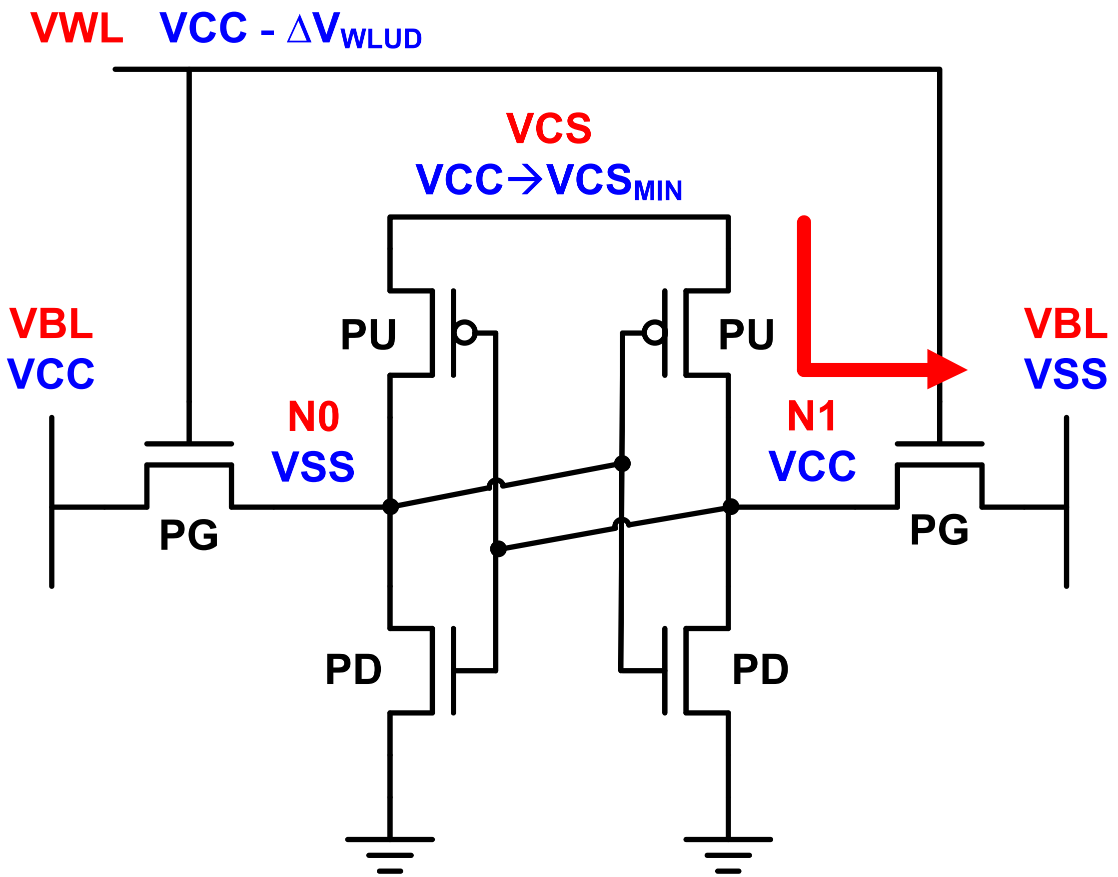
- Negative bitline and voltage collapse are the most common write assists

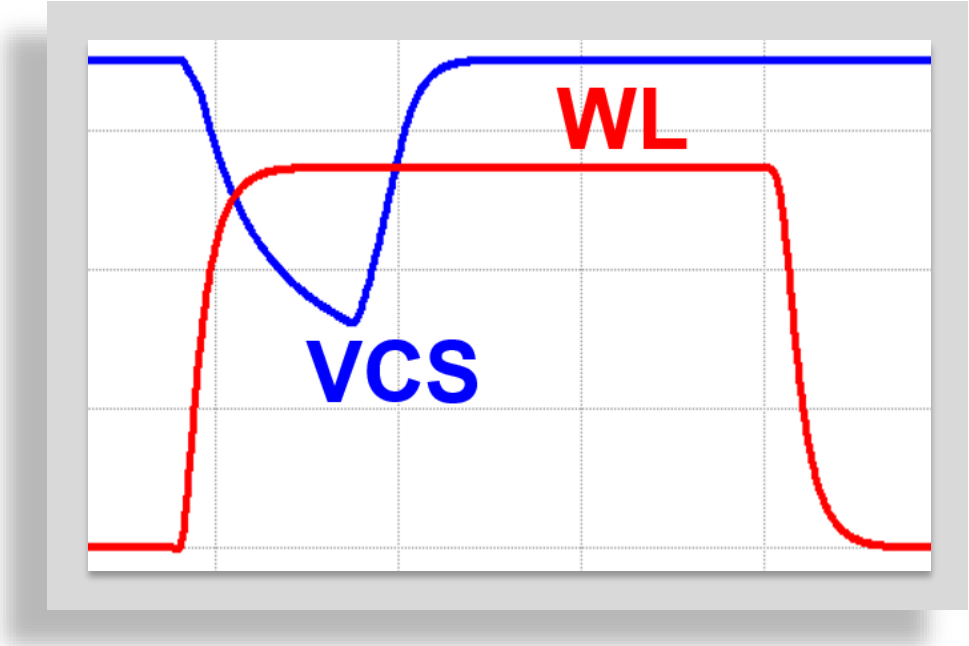
Assist	Negative Bitline	Supply Voltage Collapse	Ground Voltage Increase	Wordline Boost
Description	Coupling of low bitline below ground to increase passgate gate-source voltage	Collapse of active column cell supply to weaken latch contention	Increase of active column ground supply to weaken latch contention	Increase in wordline voltage during write to increase passgate gate-source voltage
Type	Strengthens Pass-gate	Reduces Latch Strength	Reduces Latch Strength	Strengthens Pass-gate
Adoption	Common	Common	Uncommon	Uncommon



Source: Mann, SSE 2010

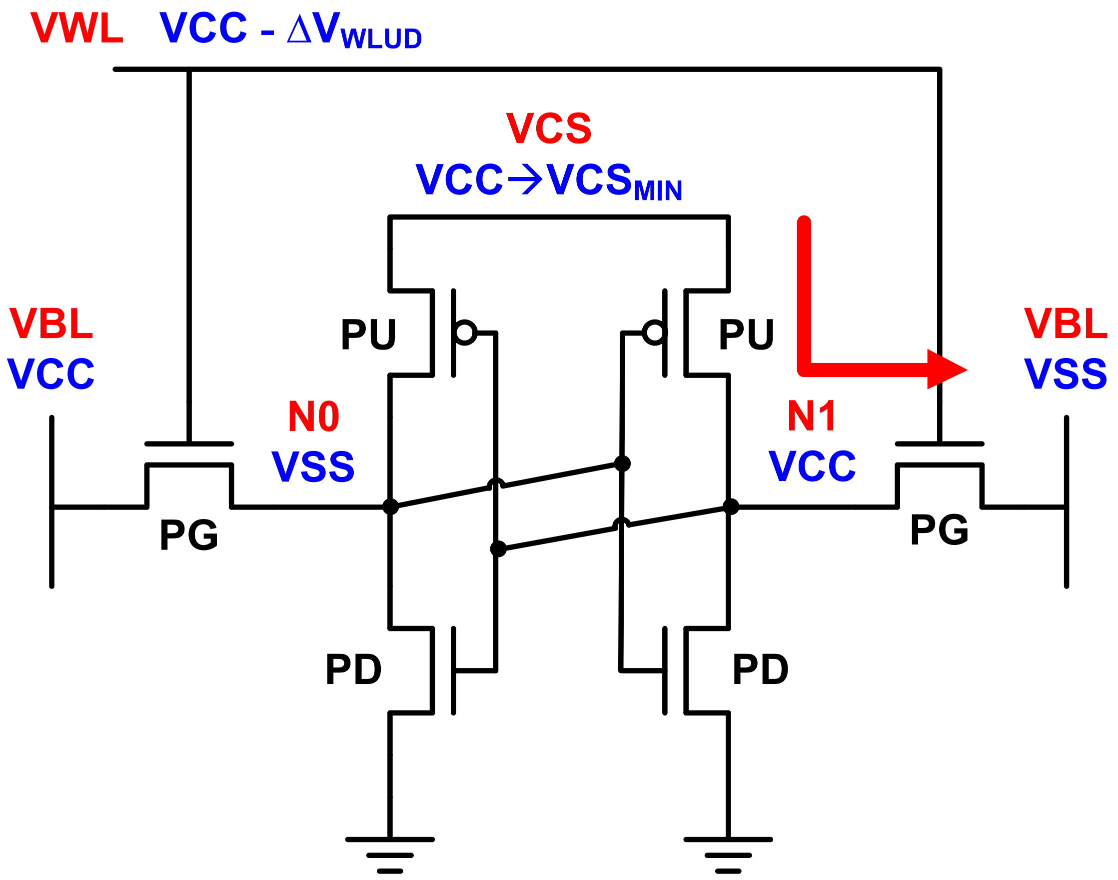
Voltage Collapse Write Assist

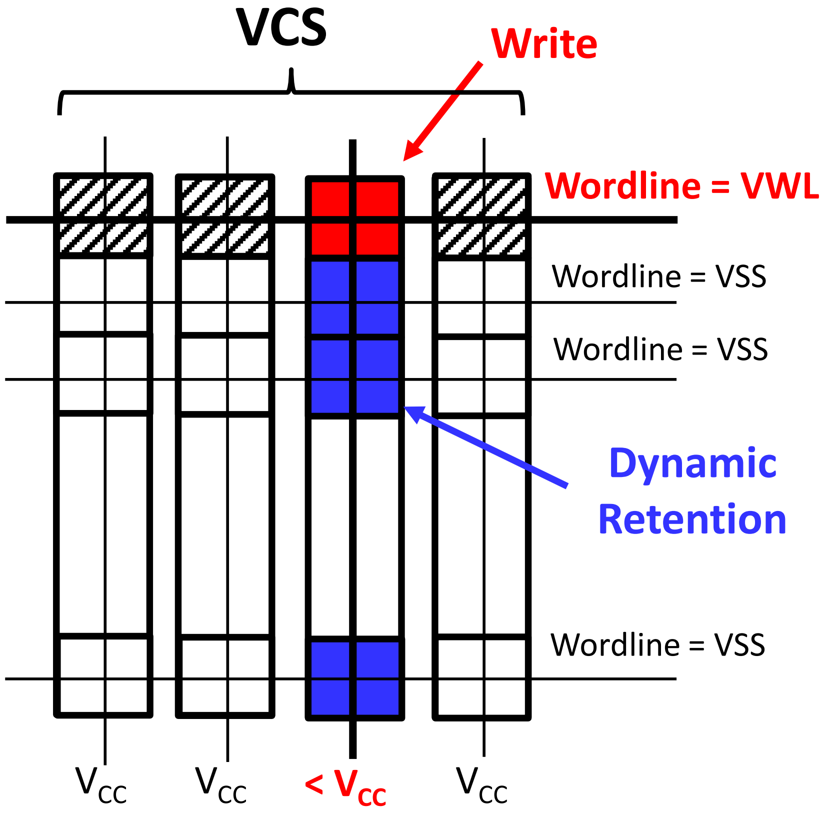




- Collapse bitcell VCC supply node to weaken PU and reduce write contention

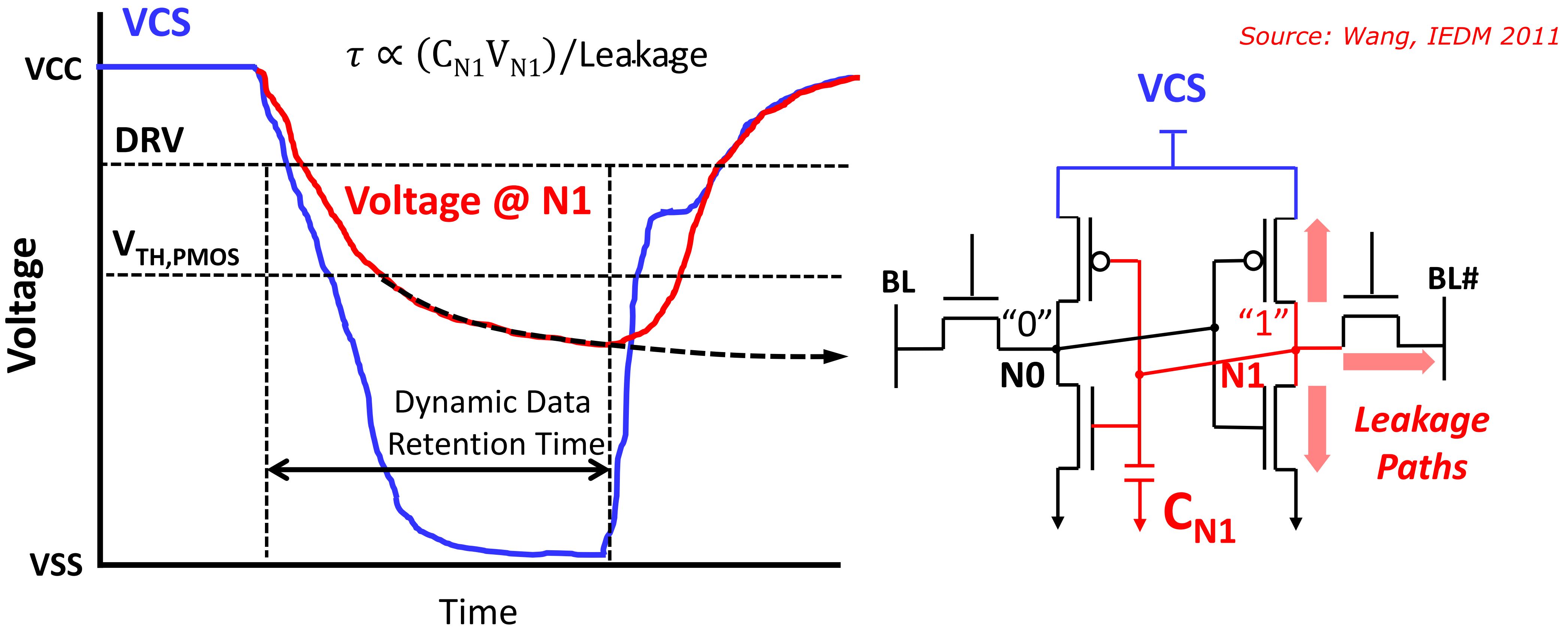
Voltage Collapse Write Assist





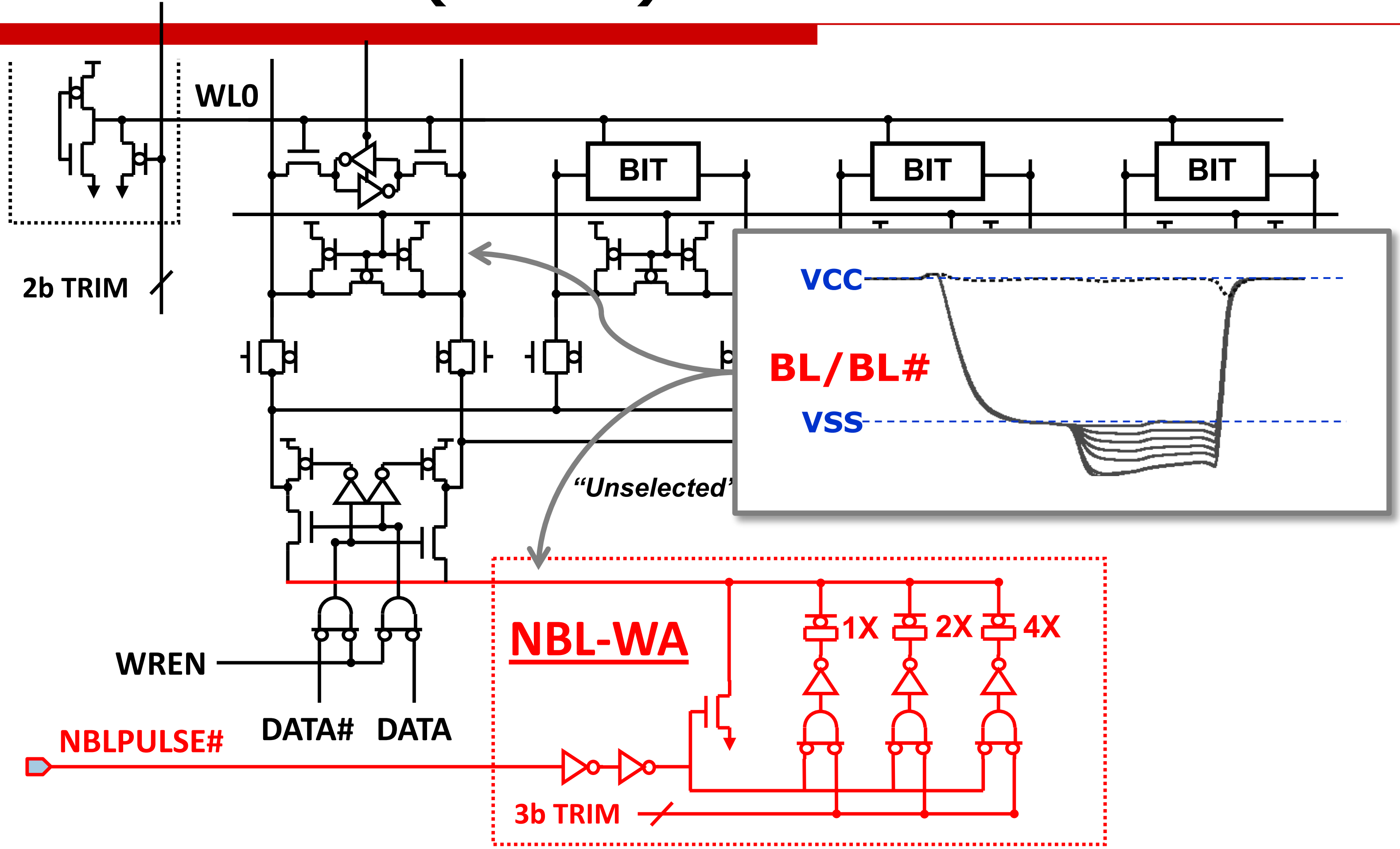
- Unselected bitcells along the column have dynamic retention risks (lose state)

Transient Voltage Collapse (TVC) Assist



- VCS can be temporarily collapsed below Data Retention Voltage (DRV)
- Timing and level are sensitive to leakage paths (transistor, defects, etc.)

Negative Bitline (NBL) Write Assist

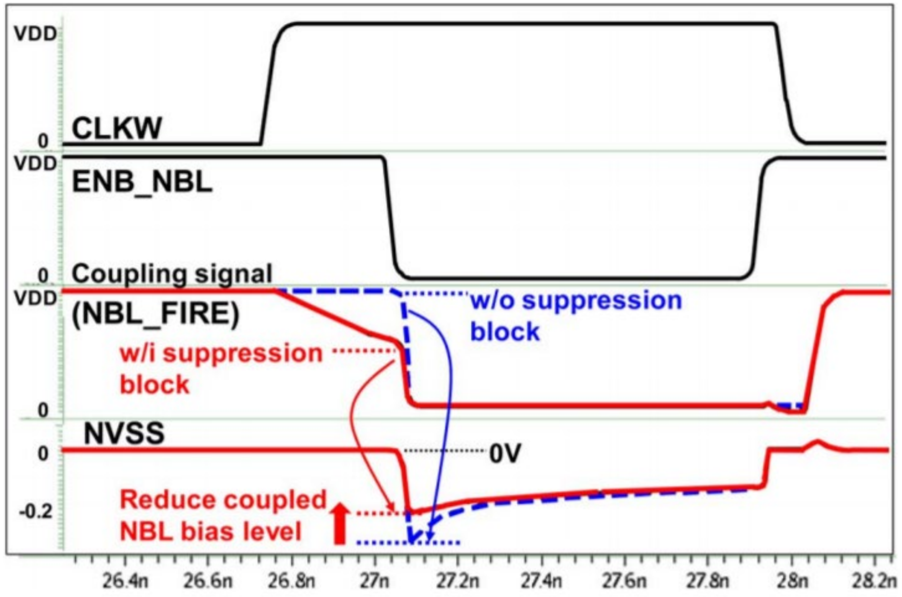


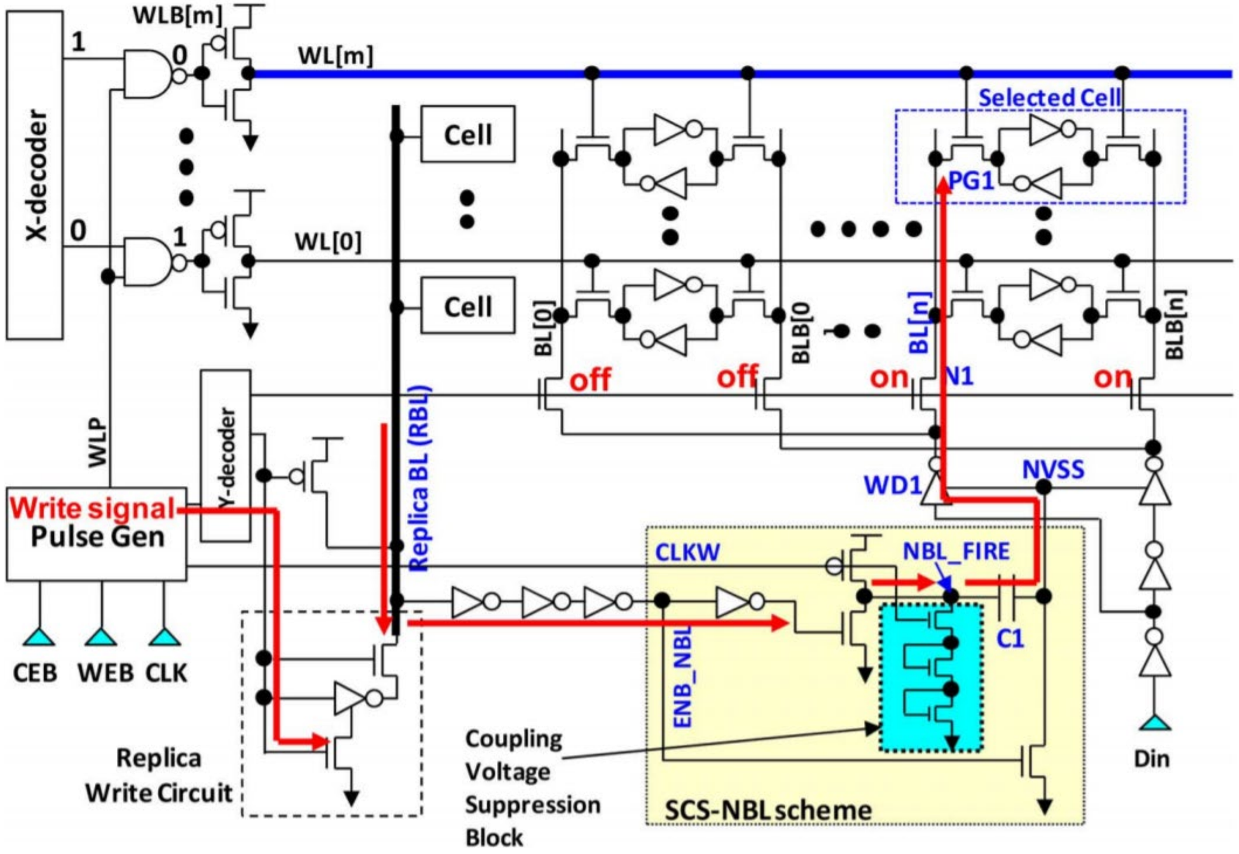
- Negative voltage on BL → strengthen passgate → break write contention

Source: Karl, IEDM 2012

NBL Write Assist Implementation

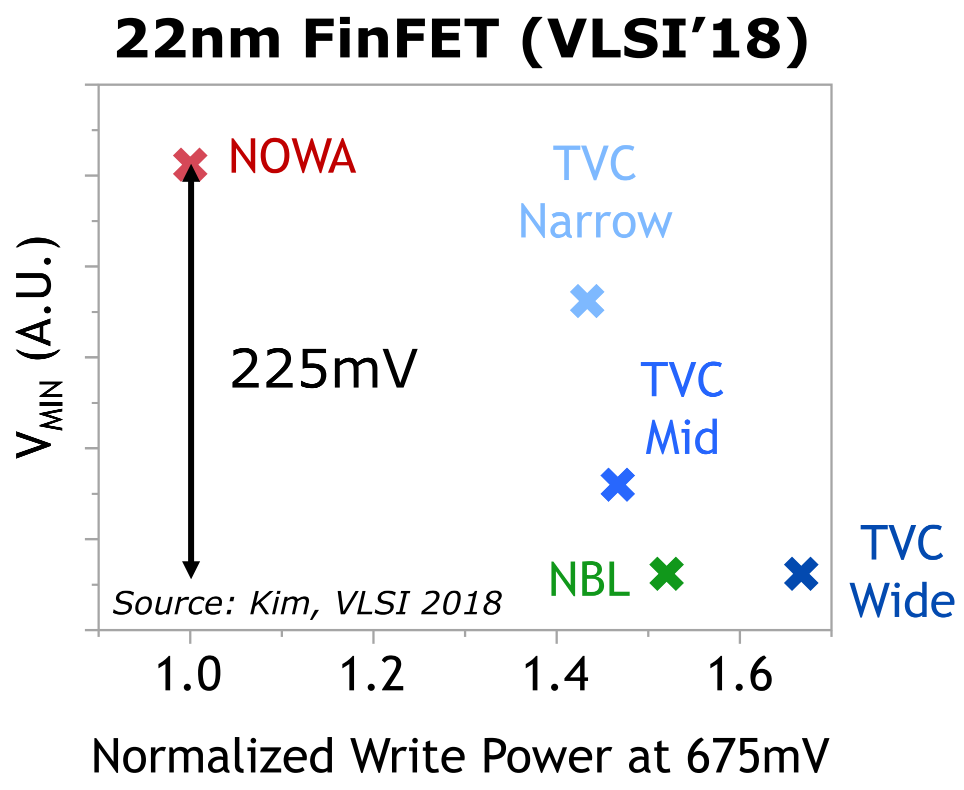
- Coupling timing tricky
  - Early → Degraded V_{MIN}
  - Late → Timing Impact
- Risk: Large negative voltage
  - Reduce charge on cap prior to coupling
  - Disable NBL at high VCC

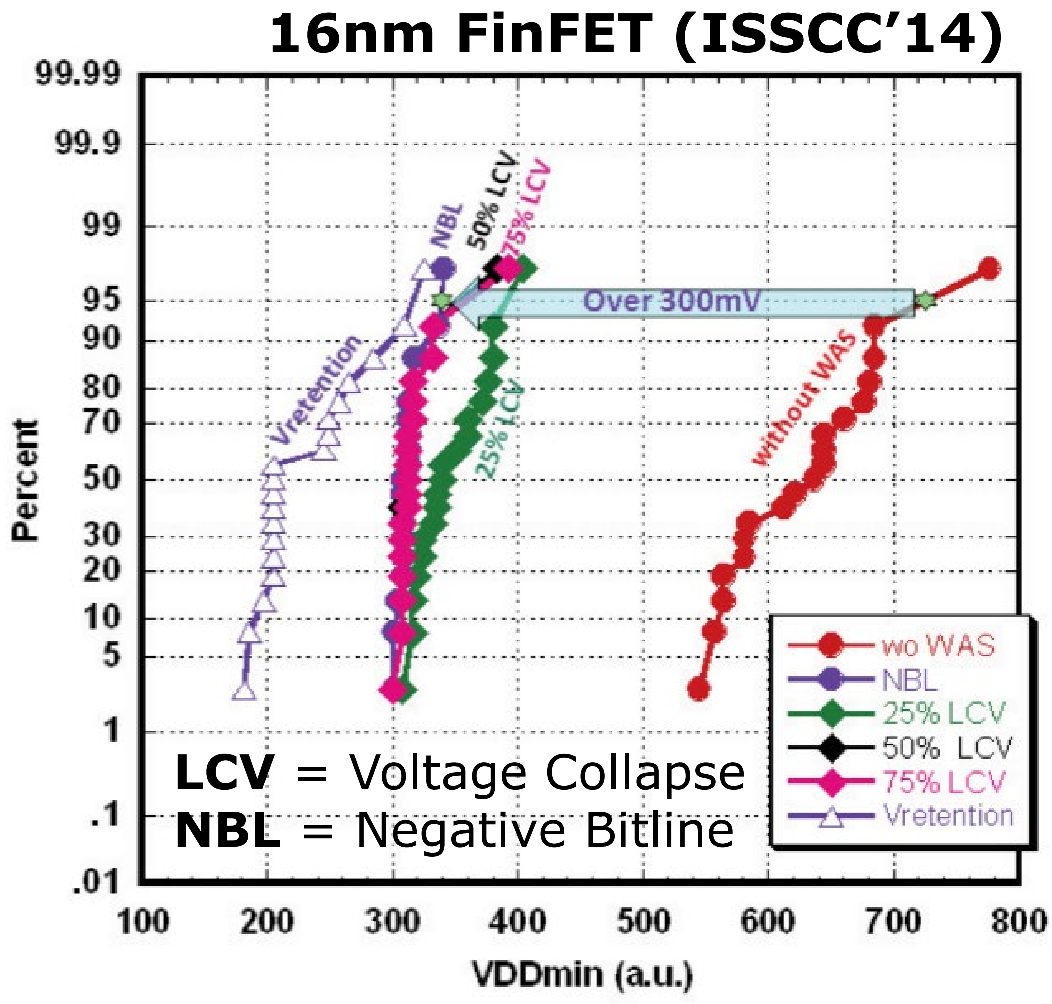




Source: Chang, ISSCC 2015

Active Write Assist Comparison





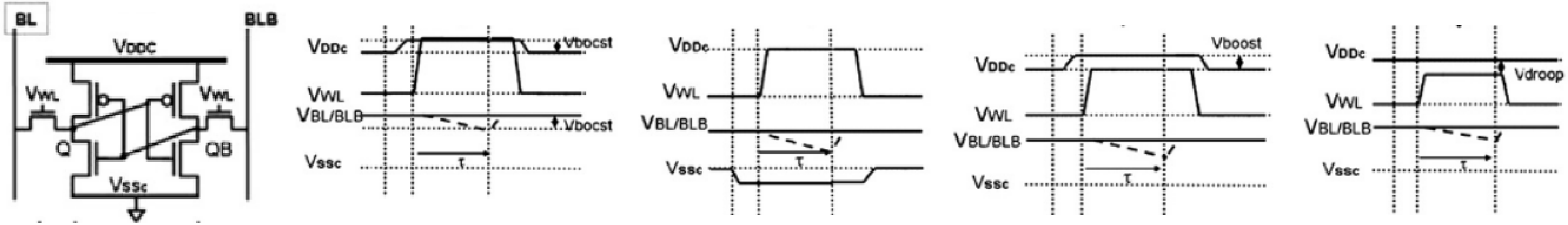
Source: Chen, ISSCC 2014

- Both TVC and NBL circuits demonstrated to deliver 200-300mV V_{MIN} enhancements on multiple nodes

Read Assist Circuits for SRAM

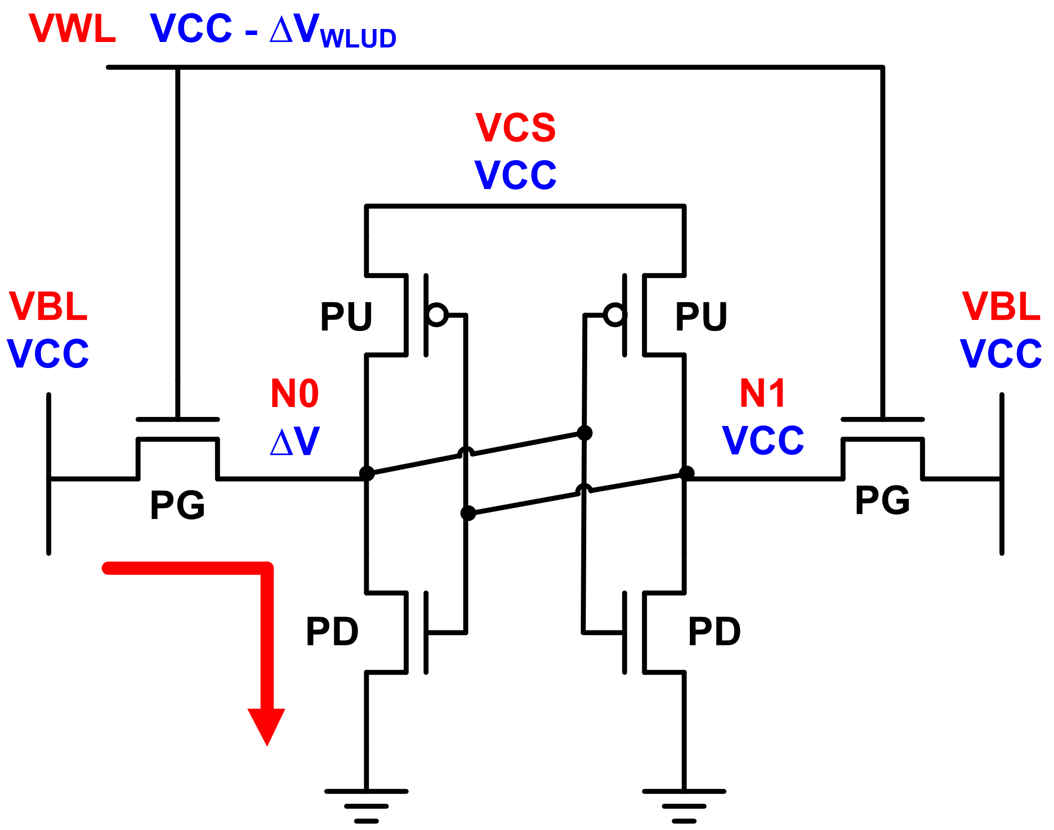
- Wordline Underdrive is the most common SRAM read assist circuit

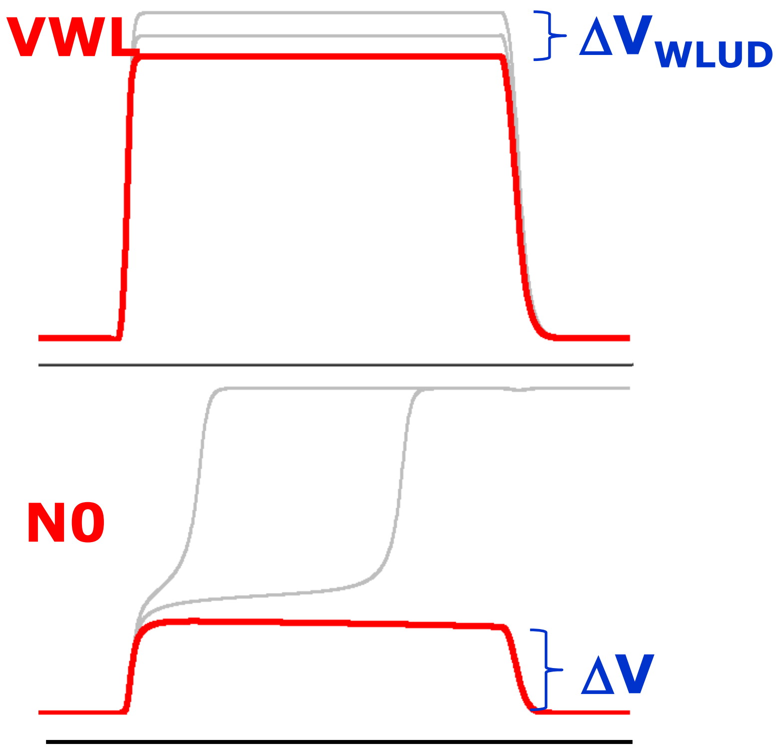
Assist	Increase Global VCC	Negative VSS at Bitcell	Boosted VCC at Bitcell	Wordline Underdrive
Description	Coupling of low bitline below ground to increase passgate gate-source voltage	Collapse of active column cell supply to weaken latch contention	Increase of active column ground supply to weaken latch contention	Increase in wordline voltage during write to increase passgate gate-source voltage
Type	Strengthens Latch	Strengthens Latch	Strengthens Latch	Reduce Noise into Latch
Adoption	Uncommon	Uncommon	Uncommon	Common



Source: Mann, SSE 2010

Wordline Underdrive (WLUD) Read Assist

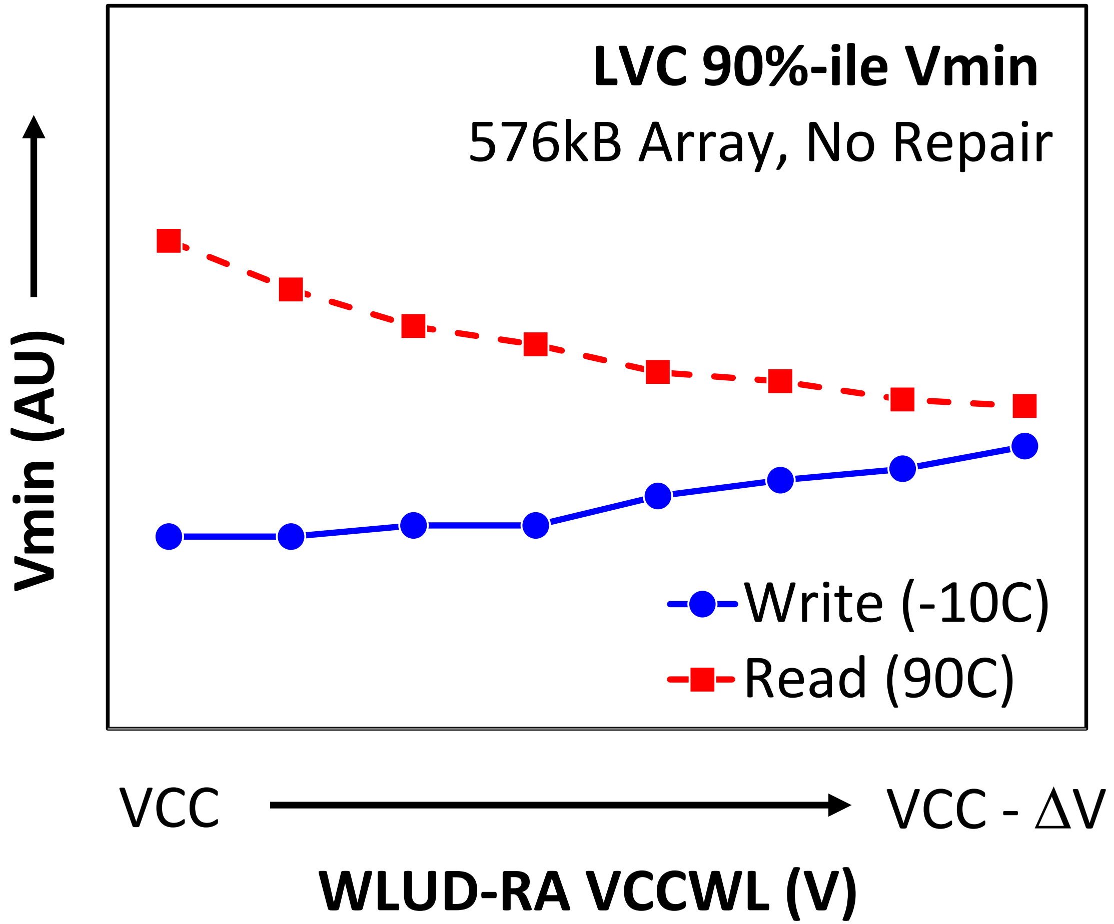




- WL voltage adjusted to tradeoff read stability vs. write margin / performance

Wordline Underdrive (WLUD) Read Assist

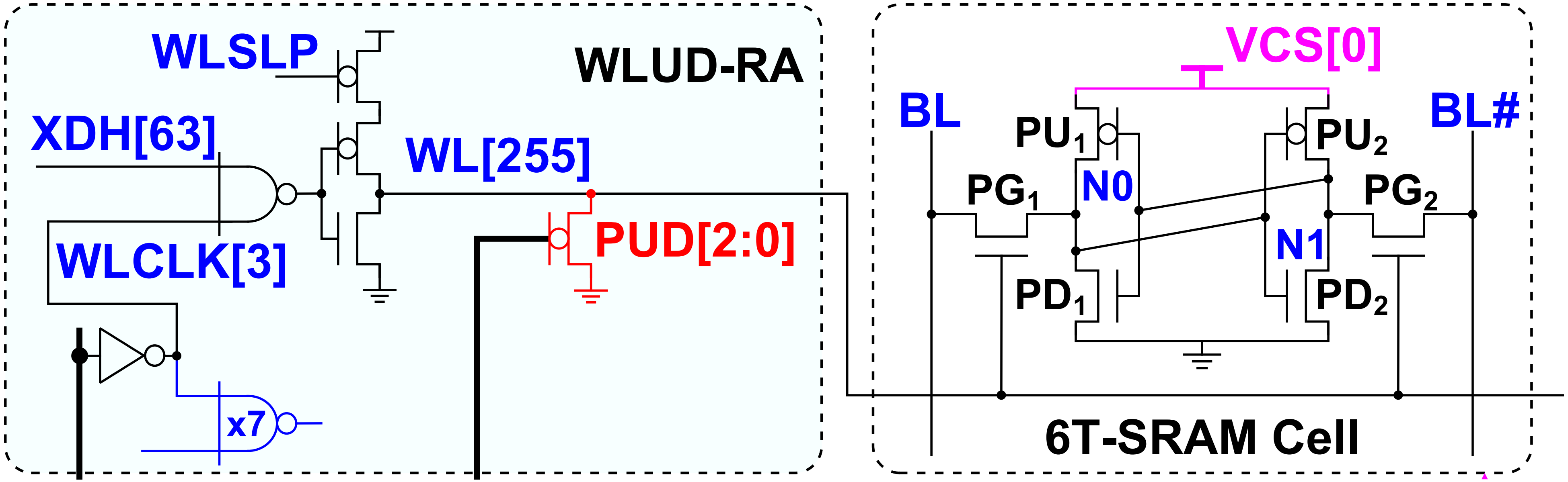
- It's a tradeoff knob → No read vs. write margin enhancement
- Usage can eliminate passgate targeting masks in some technologies
- Levels can be adjusted post-fabrication to correct for systematic variation



Source: Karl, ISSCC 2012

WLUD Read Assist Implementations

- Often implemented with simple voltage divider in SRAM row decoder driver
- Tunable pull-down NMOS or PMOS devices utilized to adjust strength
- Energy and area overhead are very modest, on the order of 1-2%



Outline

- Embedded Memory Trends
- SRAM: The Embedded Memory Workhorse
- Register Files: Special, High-Performance Memories for “XPU”
- Logic Sequentials: Ultra-Low Voltage Flip-Flops
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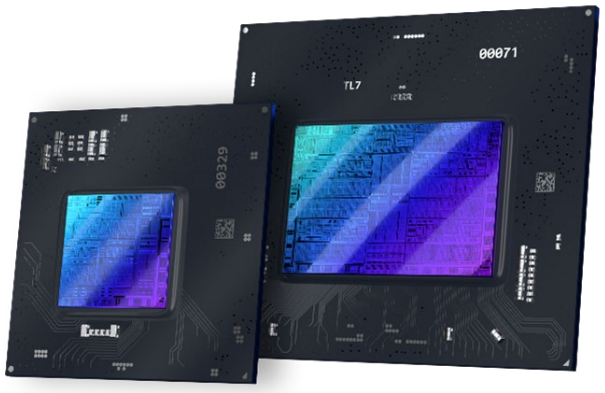
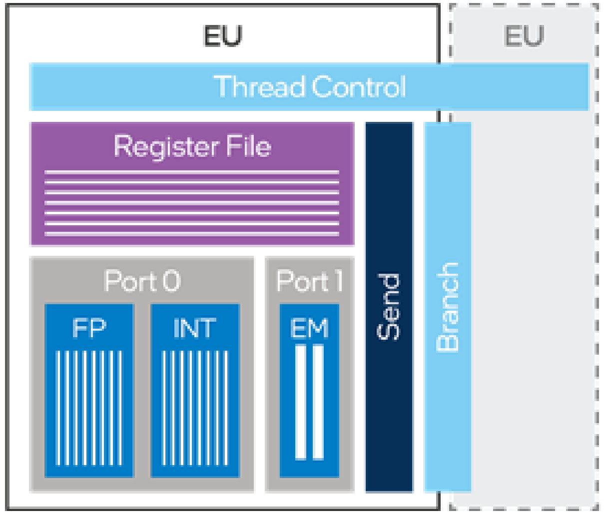
- Conclusion

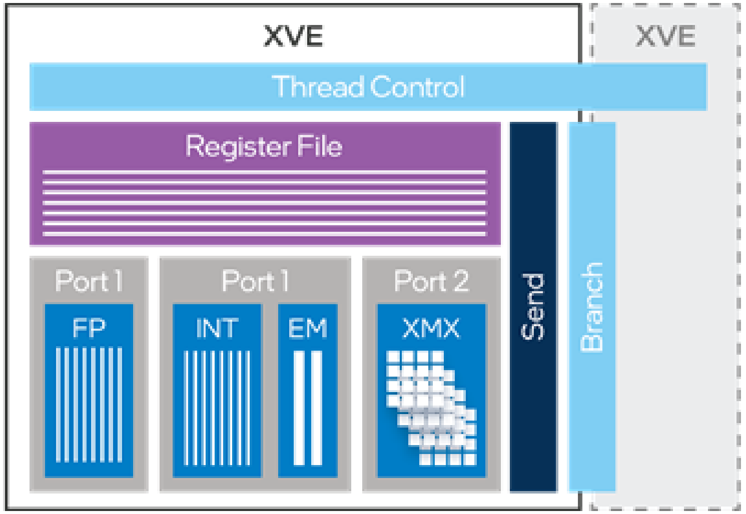
What is a Register File?

- A register file is an array of memory registers located in an CPU/GPU
- Register files are typically the lowest level of the memory hierarchy, and have **relatively high memory access rates** to support co-located execution units
- Modern register files are often implemented with **multi-ported memories** to **avoid data access conflicts** that can stall parallel execution units

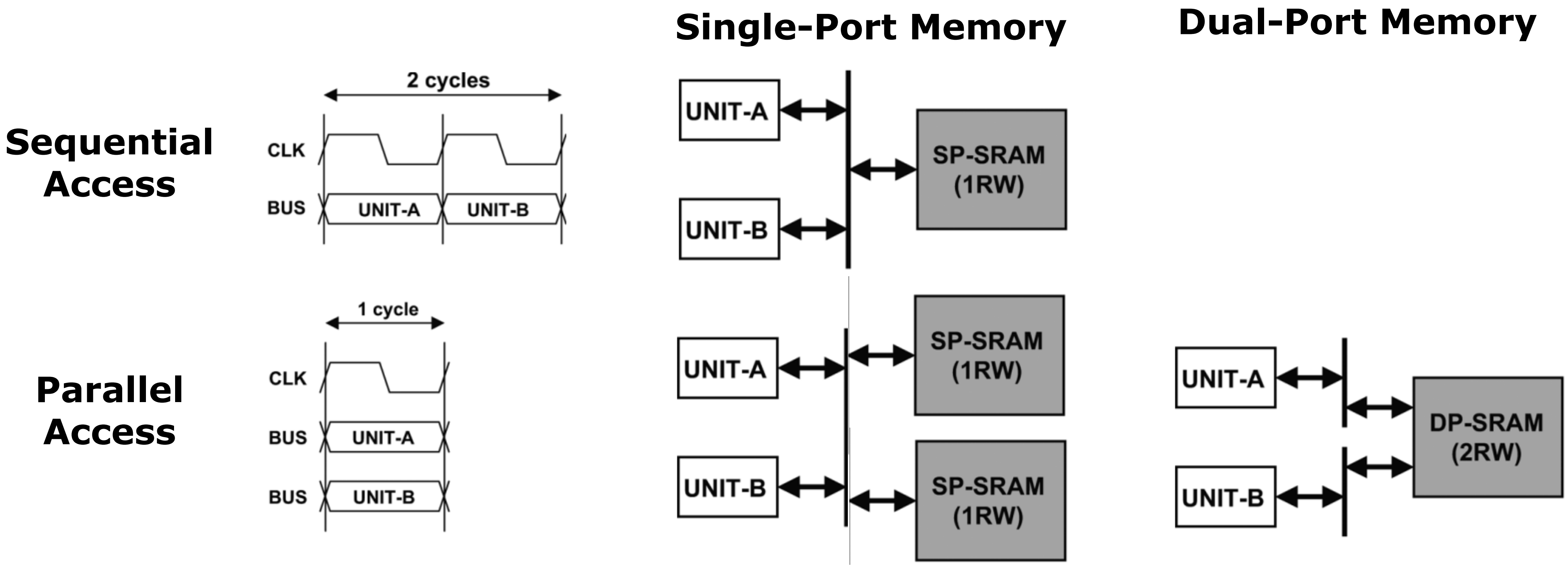


Intel Arc A-series GPU





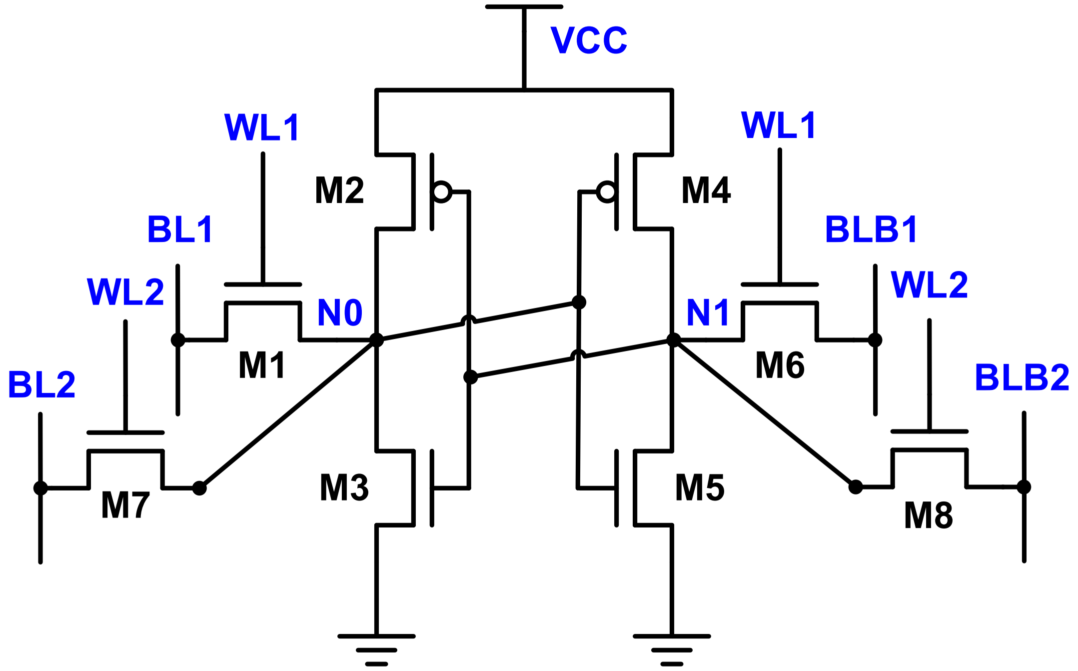
How Multi-ported Memory Helps



- Single-port memories have address access restrictions when attempting to increase read/write bandwidth per cycle
- Native, Multi-ported memories can be used to avoid access restrictions

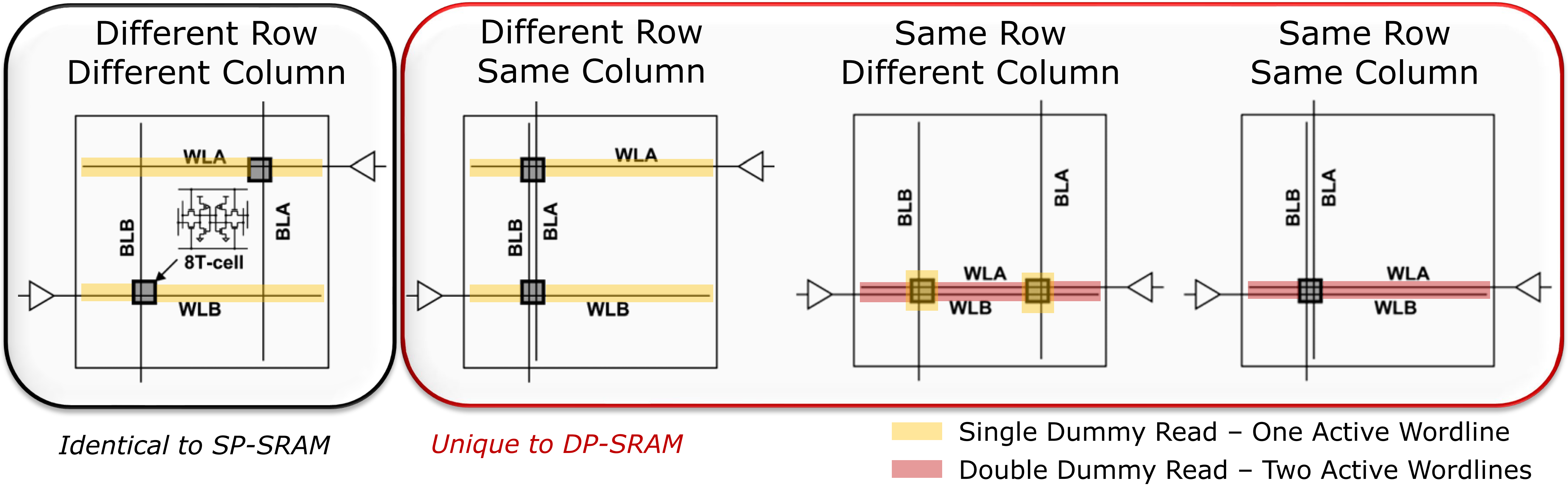
2RW Dual-Port SRAM

- Simple Idea → More passgates!
- Arrays support:
  - 2 reads concurrently
  - 2 writes concurrently
  - 1 read/1 write concurrently
- Requires stronger M3/M5 → additional noise injection from M1/M6/M7/M8 devices during reads



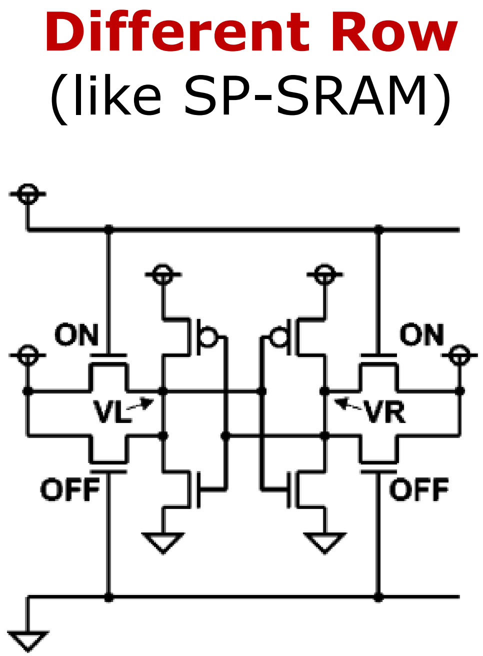
- "True" 2RW dual-port SRAM provides the most flexible access characteristics for a 2-port memory, but introduces unique margin and timing challenges

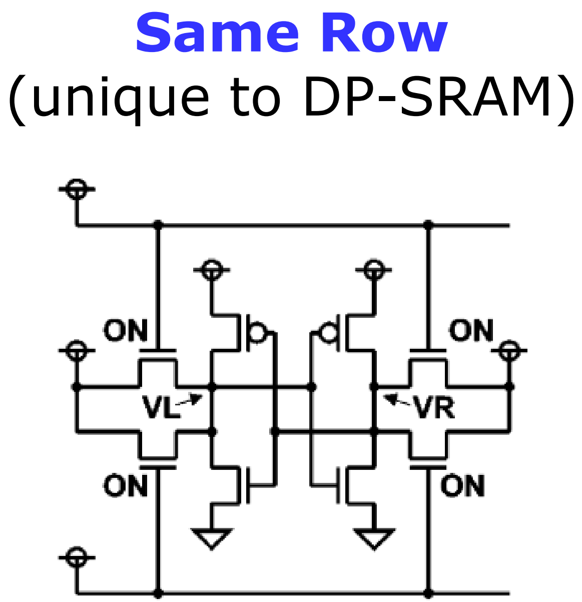
2RW DP-SRAM: Supported Array Accesses

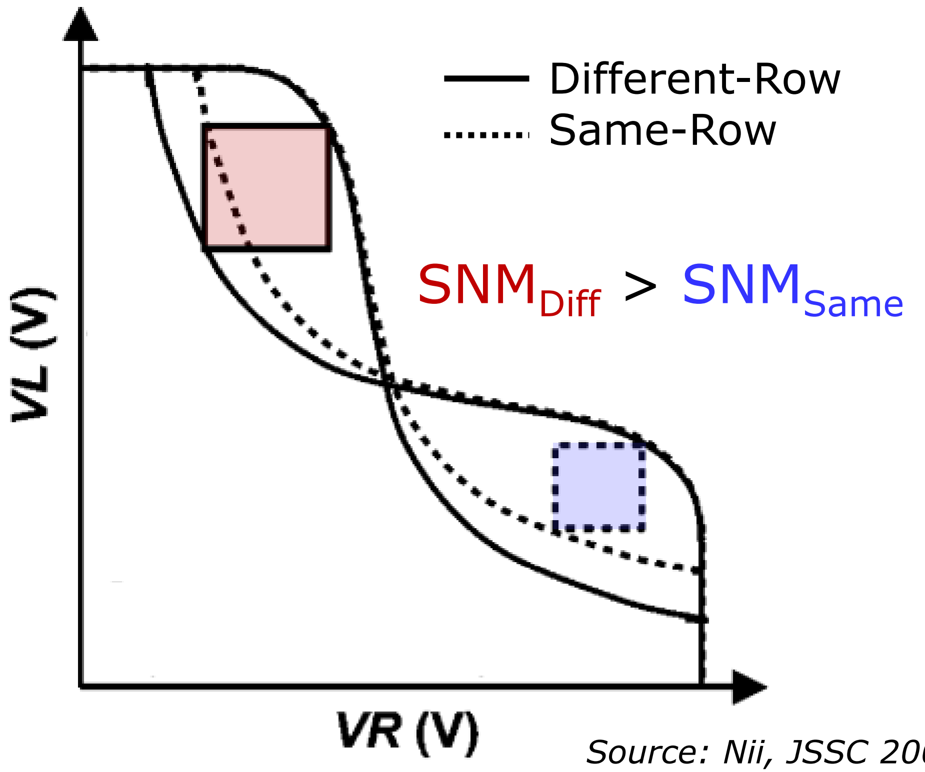


- Same-row access introduces "Double Dummy Read" in un-accessed cells
- Same-row different column introduces "Single Dummy Read" interacting with read/write operations in the selected cells
- Write on both ports to the same cell is not commonly supported

2RW DP-SRAM: Bitcell Stability



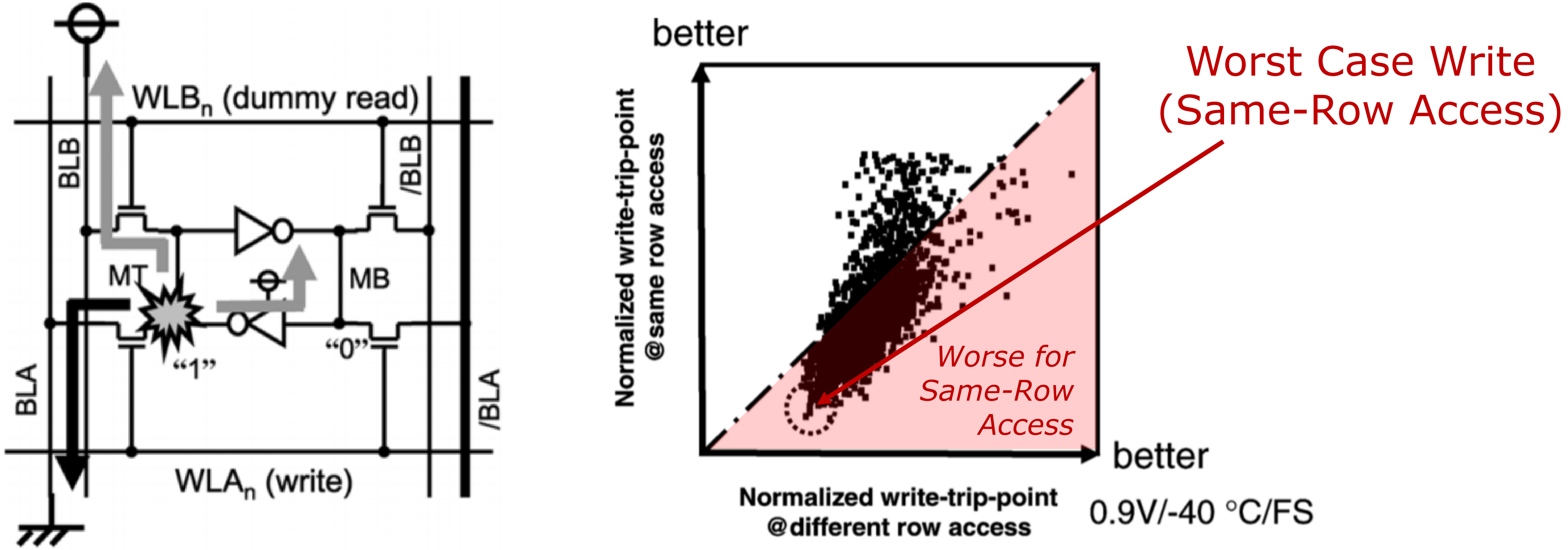




Source: Nii, JSSC 2009

- Same-row accesses introduce concurrent read or dummy read operations
- This reduces the noise margin (cell read stability) and requires a larger bitcell design (larger NMOS PD) or external circuit assists to stabilize the cell

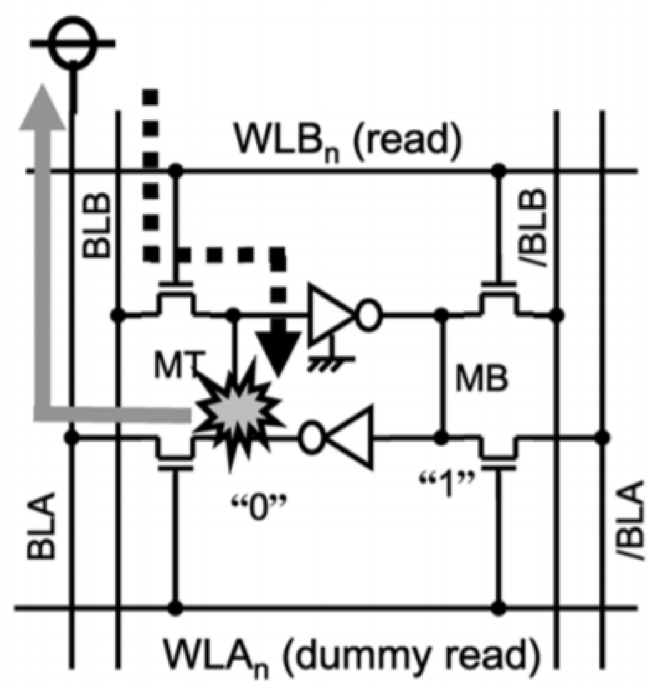
2RW DP-SRAM: Read Disturb Write

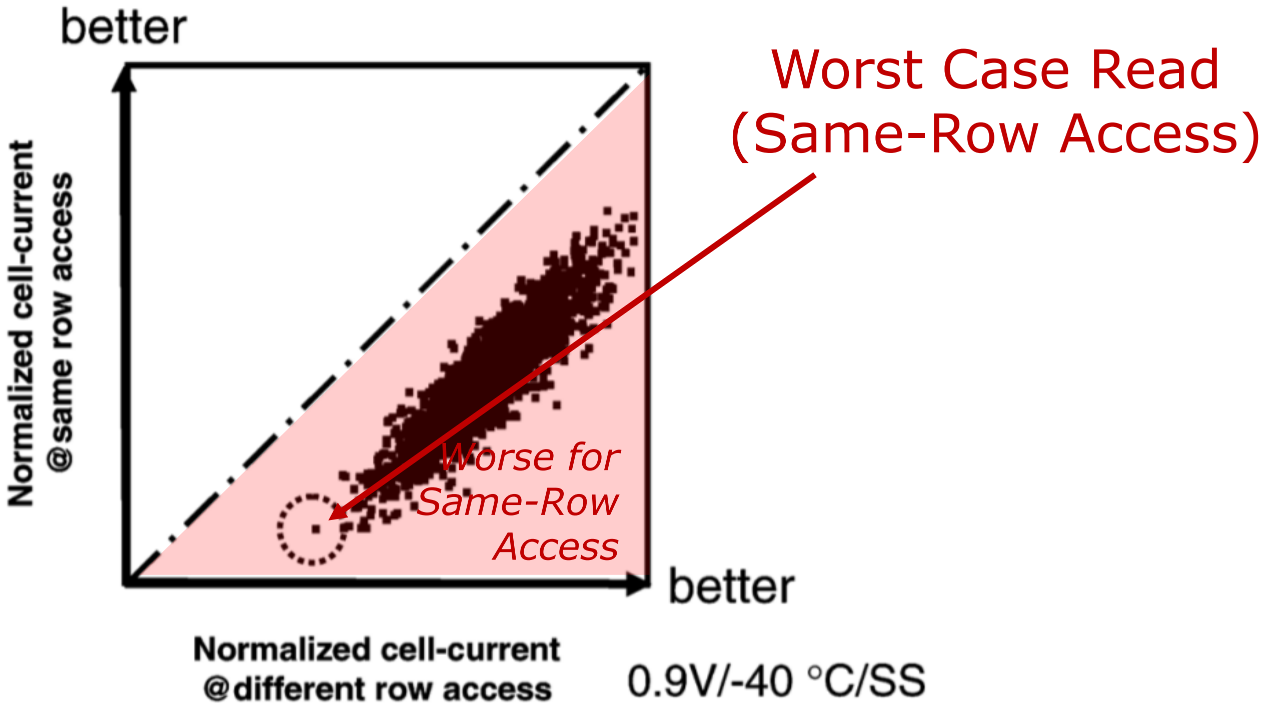


- Dummy read creates additional contention with write operation
- Dummy read bitline is commonly held in precharge state (at VCC) to avoid interfering with write operations on other ports

Source: Ishii, JSSC 2011

2RW DP-SRAM: Read Disturb Read

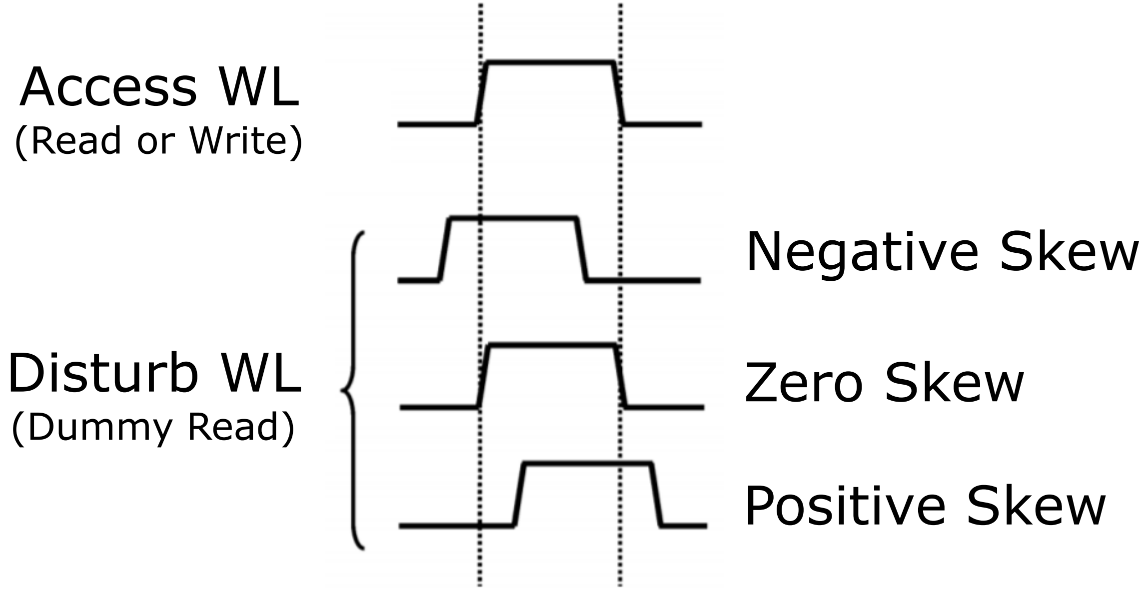




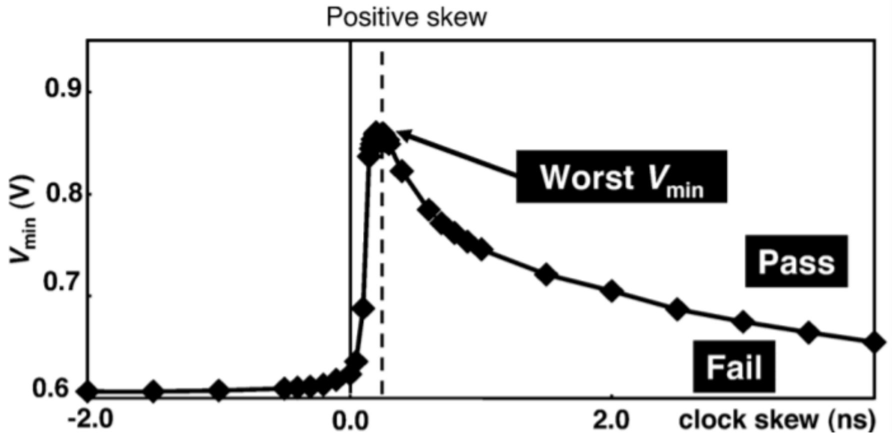
- Same-row access also sets the worst case read current
- Dummy read colliding with read operation in the same row, leads to the lowest read current (since dummy BLs are held at VCC)

Source: Ishii, JSSC 2011

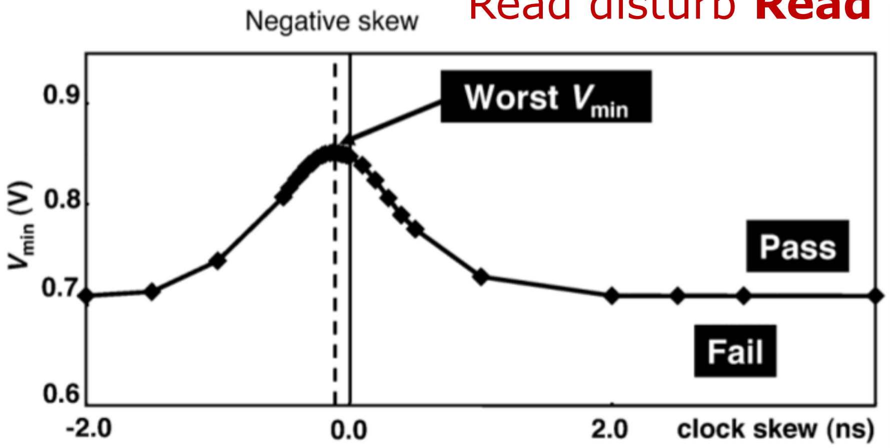
2RW DP-SRAM: WL Skew Sensitivity



Read disturb **Write**



Read disturb **Read**

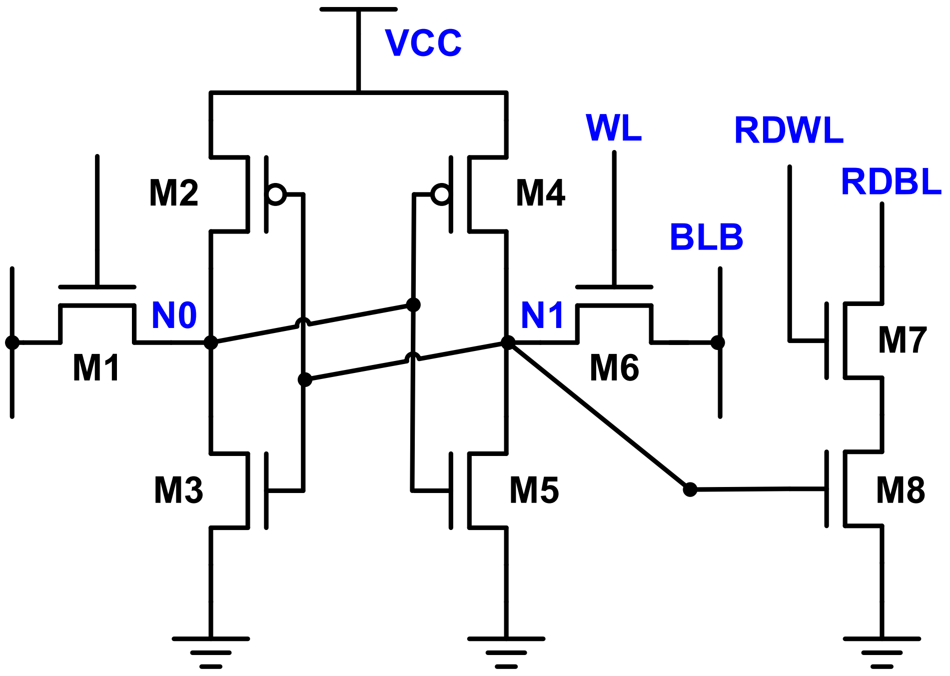


- DP-SRAM ports are often clocked independently (asynchronous access)
- Timing alignment of disturb WL to access WL in Same-Row access is critical to identify worst case timing and cell margins

Source: Ishii, JSSC 2011

1R1W Dual-Port 8T SRAM

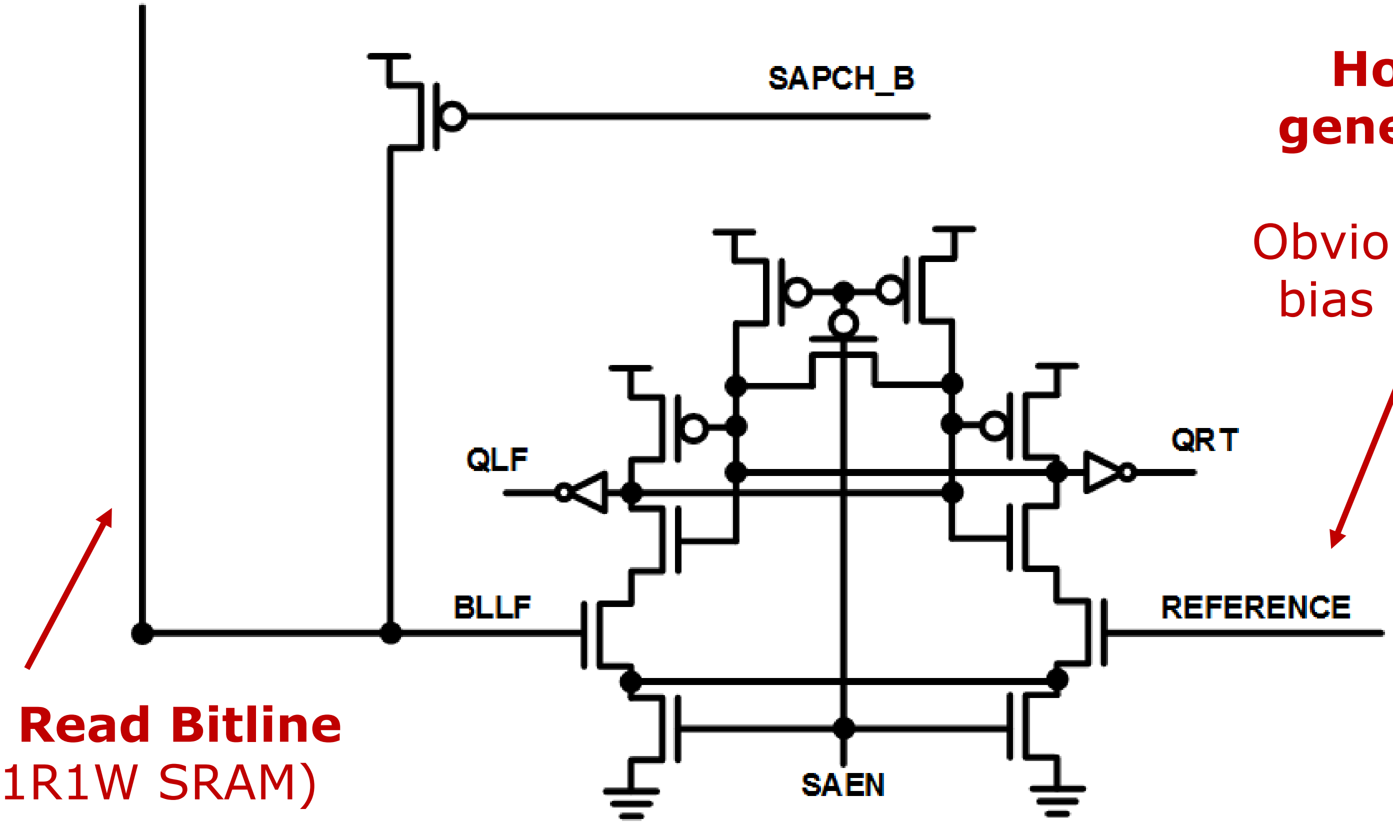
- 6T SRAM + **Dedicated Read Port**
- Arrays can support:
  - 1 read/1 write concurrently
  - 2 reads concurrently (less common)
- Common reuse of 6T SRAM bitcell design in the core of this 8T SRAM
- Simpler array IP design and technology margining due to fewer unique access conditions
- Single-ended read port → different tradeoffs for read performance



- 1R1W dual-port SRAM doesn't support two concurrent write operations, but has fewer timing and margin challenges than 2RW dual-port

Single-Ended Small Signal Sensing

- Primary difference between 1R1W 8T SRAM and 1RW 6T SRAM is implementing read sensing scheme without differential bitlines



Single Read Bitline
(with 1R1W SRAM)

How are we going to generate this reference?

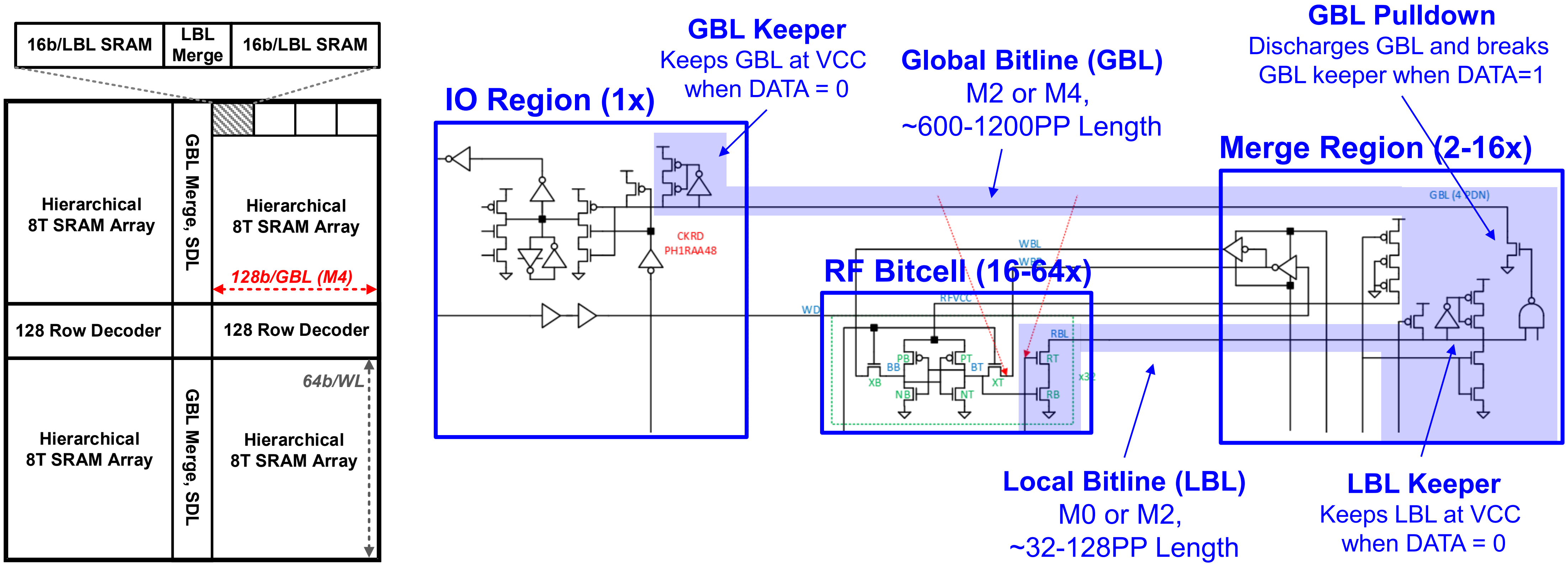
Obviously, an external analog bias generator is an option.

Can you keep the analog bias noise free routing across the memory?

Can you afford the area/power to integrate analog bias generation closer to the memory?

Large Signal Hierarchical Sensing

- Hierarchical local-global bitlines can be utilized with wide-OR domino circuits to for power-efficient sensing of 1R1W read port
- Keeper circuit contention across interconnect is key technical challenge



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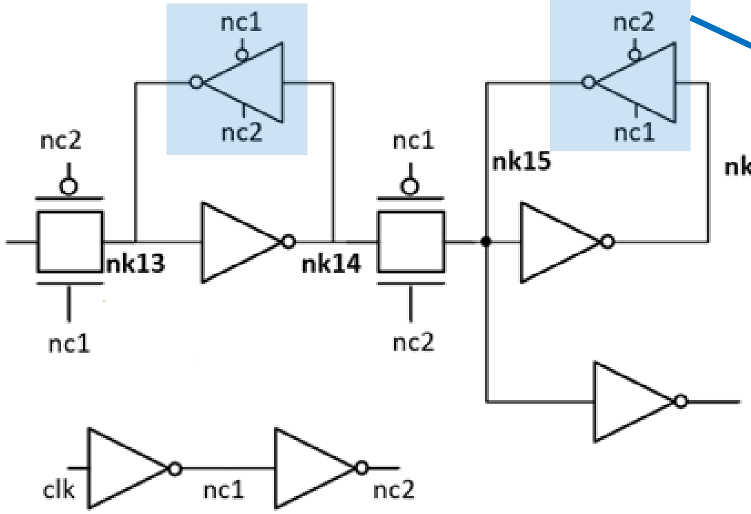
Logic Sequentials as Memory

- Traditional, compact SRAM and multiported memory relies upon ratioed transistors to ensure functionality
  - This is a fundamental limiter to reaching lowest operating voltages!
- Interruptible Logic Sequentials can be used as memory to take you to even lower voltage operating points



Primary-Secondary Flip-Flop

Fully Interruptible Feedback on Latch Elements

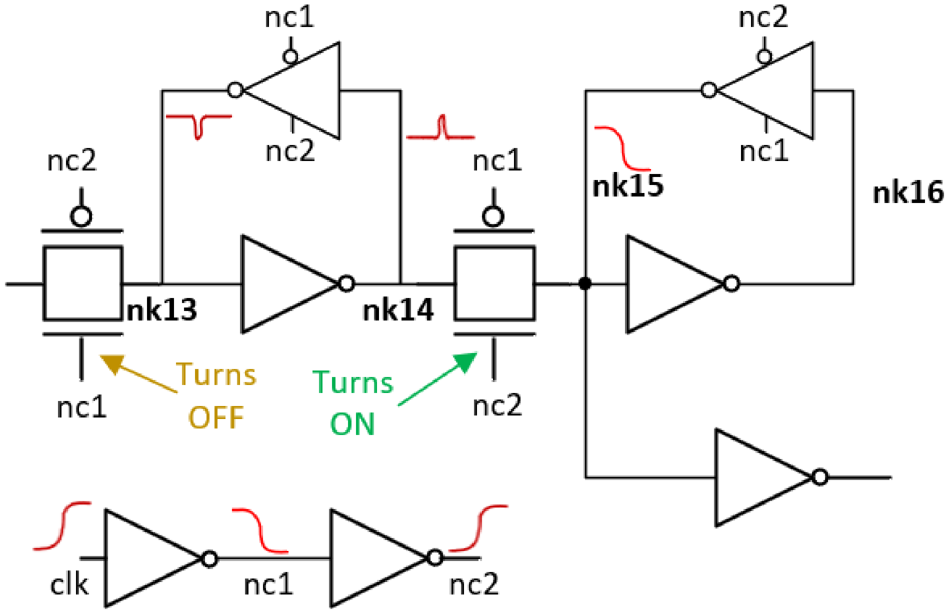
Eliminates circuit functionality dependence on ratio between transistors, enabling more reliable operation at low voltage

Vmin Failures in Logic Sequentials

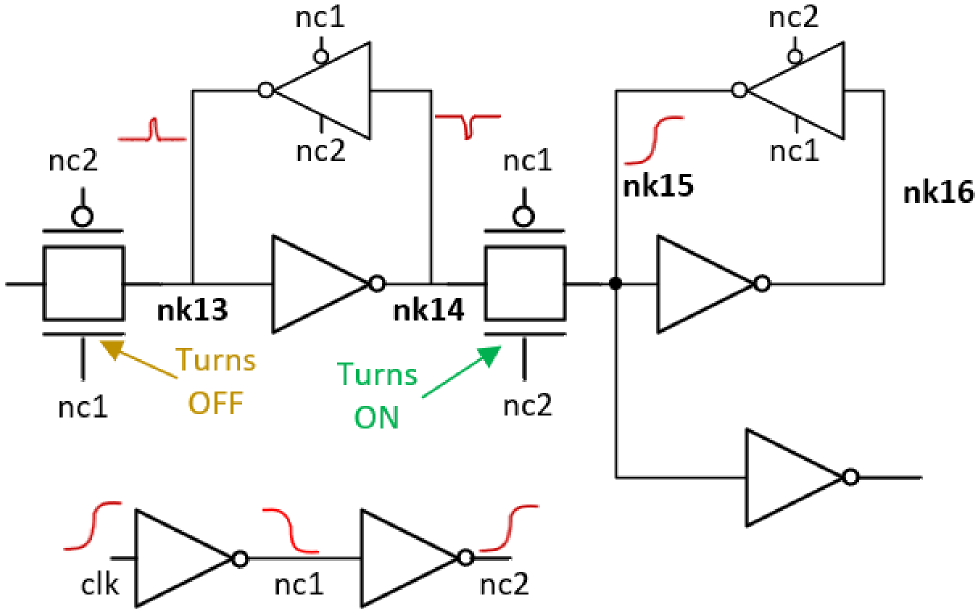
- Even fully interruptible state elements can fail at lower voltages, depending upon the circuit topology employed
- For the primary-secondary flip-flop from our example, the typical limiters at lower voltages include:
  - Write-back charge-sharing across the pass-gate
  - Internal min-delay failures related to internal clock slopes

	Standby Vmin	Active Vmin			
Failure Mode	Retention	Write-Back	Write-Disturb	Internal Hold	Scan-Stitch
Description	Lowest static retention voltage under variation	Charge-sharing glitch from secondary latch disturbs primary latch state	Excessive keeper leakage disturbs primary/secondary latch state when keeper loop closes	Min-delay failure between primary and secondary latch	Min-delay failure between bits in internally stitched multi-bit FF scan chain
		Common Limiter		Common Limiter	

Write-back Failure Mechanism



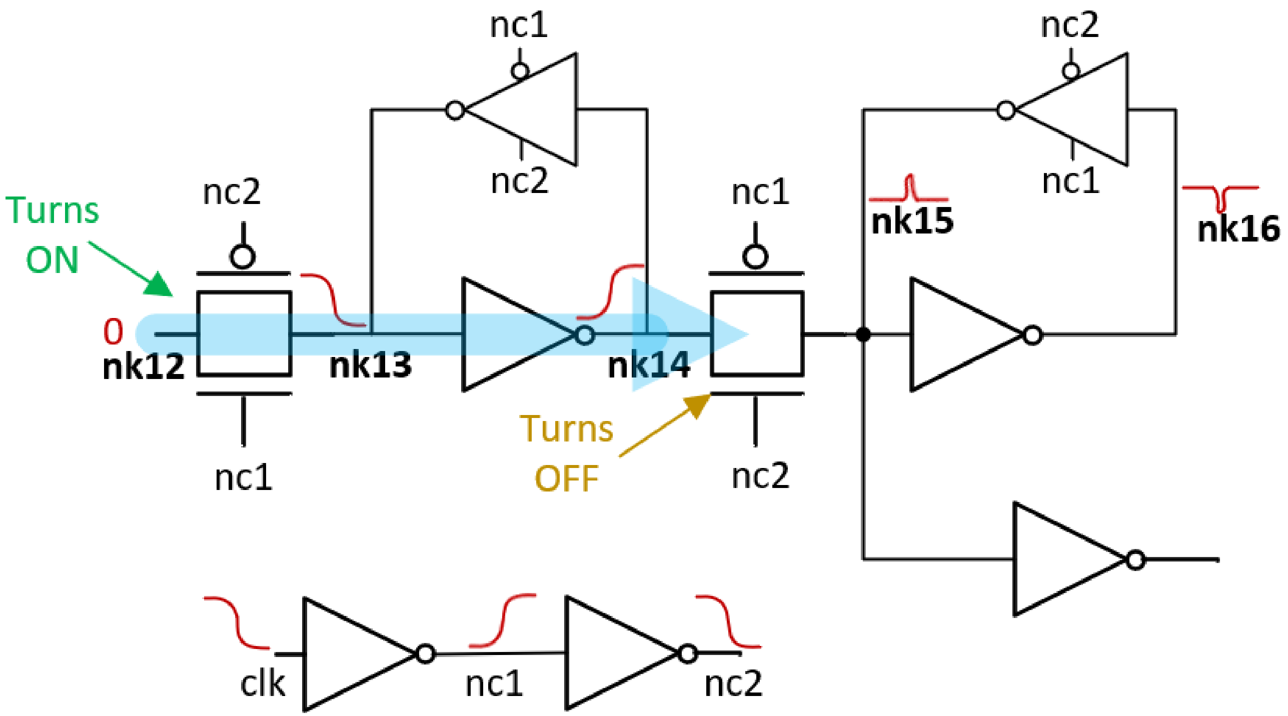
WB- high



WB- low

- Momentary charge-sharing from nk15 node to nk14 can compromise primary latch before clock transitions are complete
- Typically failing first on higher threshold voltage flip-flops, at low temperature

Internal Min-Delay Failure



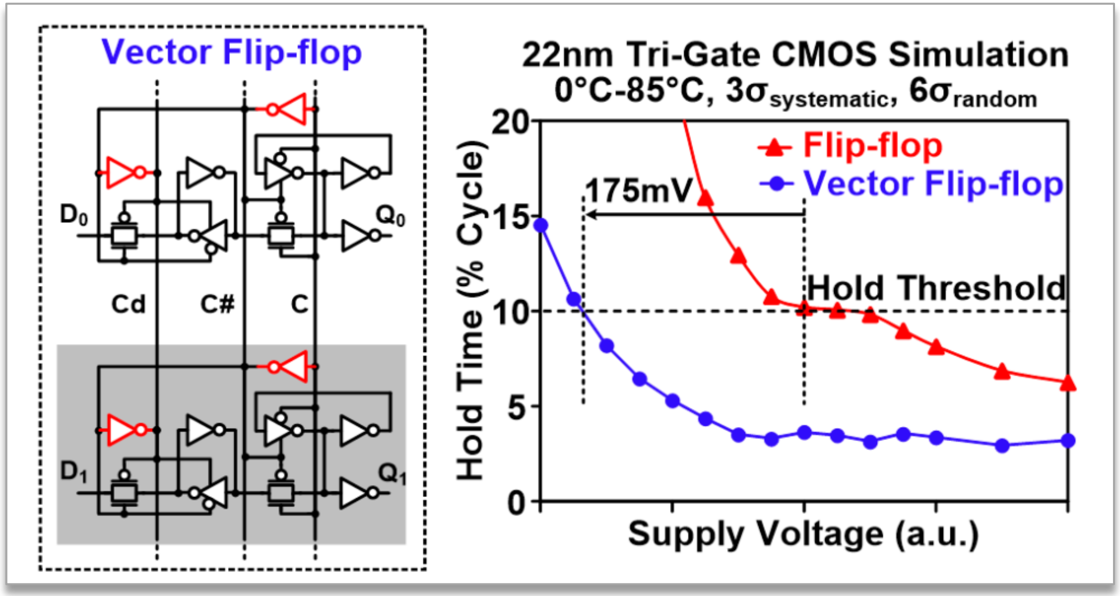
- Glitches through internal passgates during clock transitions also start to cause min-delay failure at secondary latch
- Typically failing first on lower threshold voltage flip-flops, at high temperature

Clock Optimization at Ultra Low Voltage

Mind the clocks!

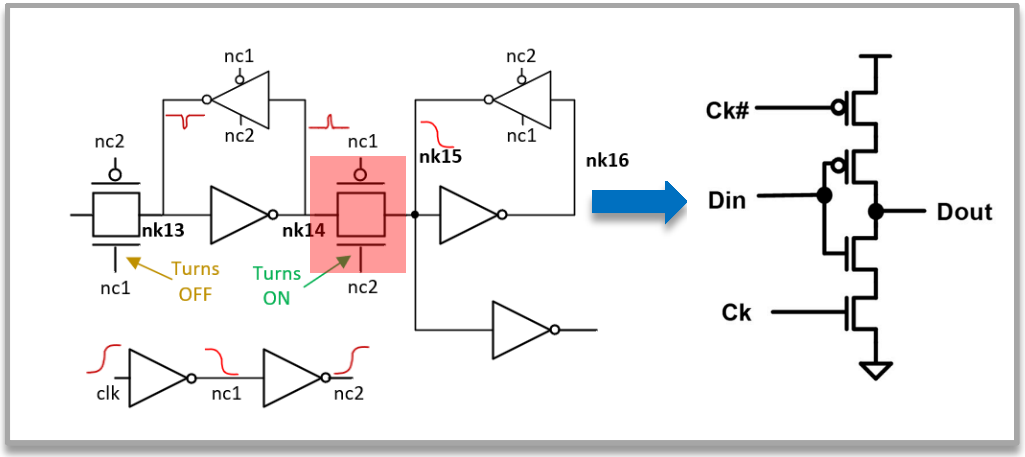
All logic sequential failure modes are impacted by clock signal degradation

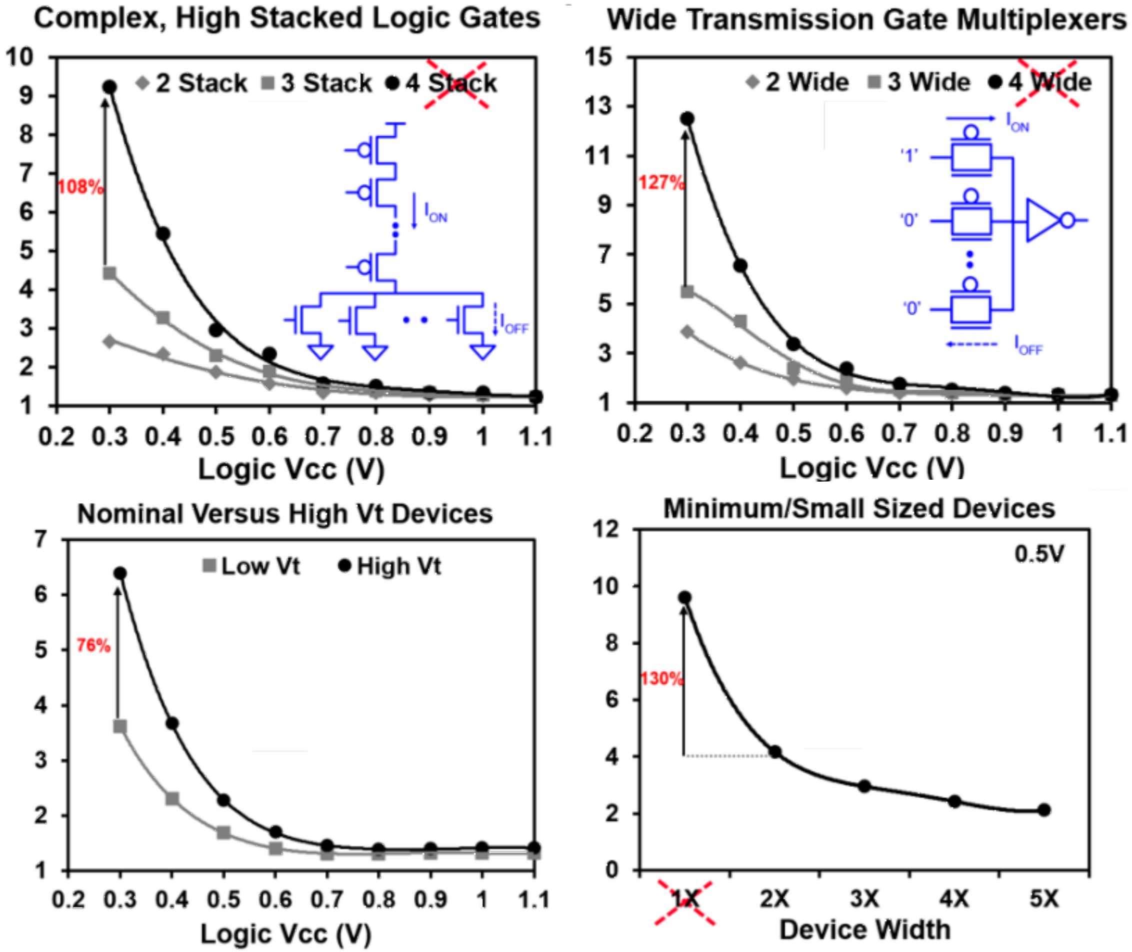
- Several techniques can extend to lower voltage operation:
  - Local clock driver upsizing and Vth mapping optimization
  - Local clock layout optimization
  - Large Vector flop value grows due to amortizing upsized local clocking



Circuit Topologies for Ultra-low Voltage

- Circuit topology adjustments and restrictions enable lower V_{MIN}:
  - Stacked Device Restrictions
  - High Vth Device Usage Restriction
  - Small Device Usage Restrictions
  - Transmission Gate Avoidance
- Results in tradeoffs to area, leakage and capacitance





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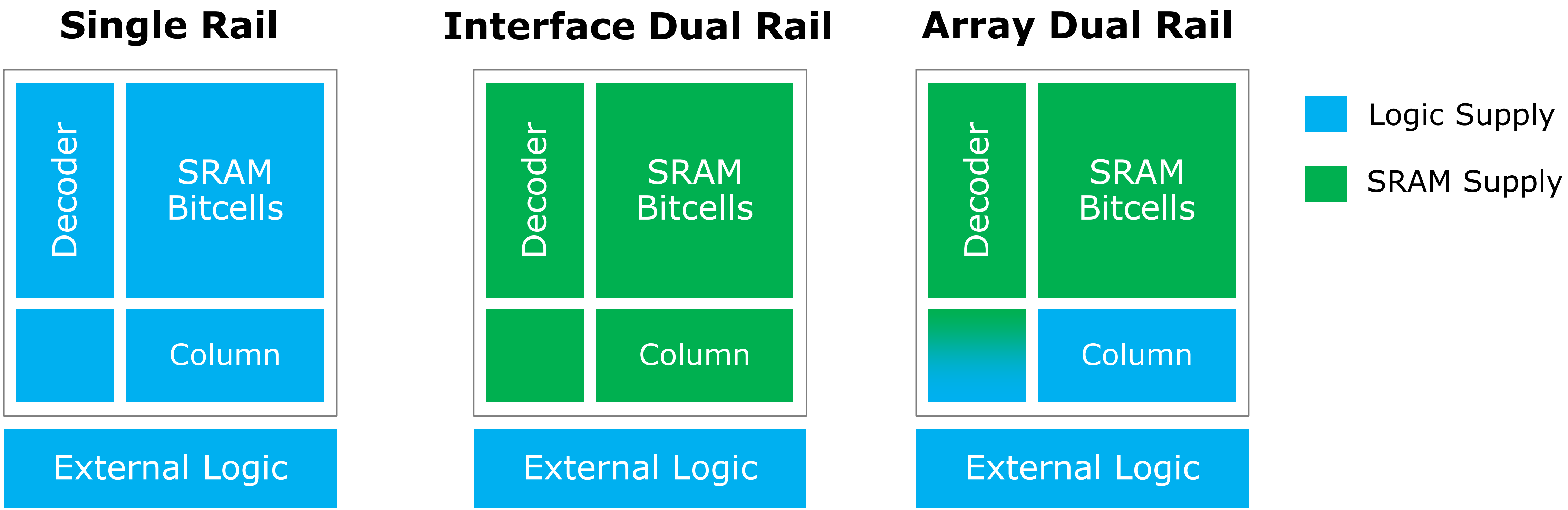
Supply Voltage Strategies



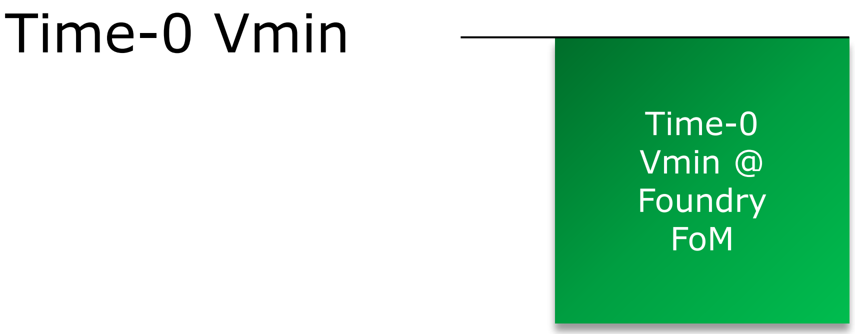
Single Rail
- Simple, Easy Integration
- Memory can limit Logic operating voltages

Interface Dual Rail
- Allows wide flexibility between Logic/SRAM supply voltages

Array Dual Rail
- Lower array energy
- Supply delta limitations between SRAM/Logic

System Vmin with Single Rail Memory

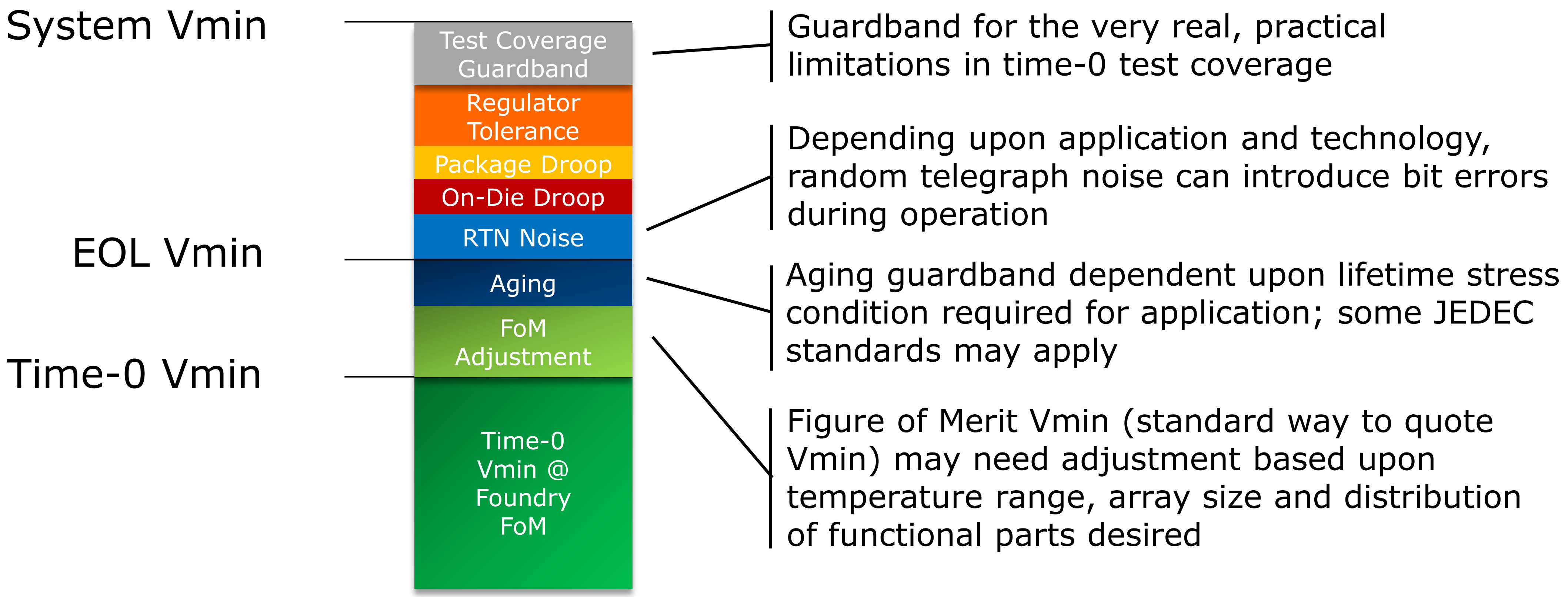
- "Single Rail" design solutions make sense in applications pursuing energy efficiency with high access rates to memory



Will this memory work on my existing system supply voltage rail?

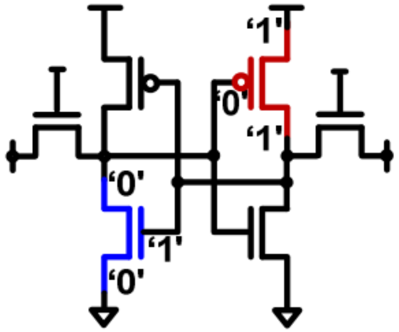
System Vmin: Some Assembly Required

- Care is required to translate memory Vmin to system Vmin

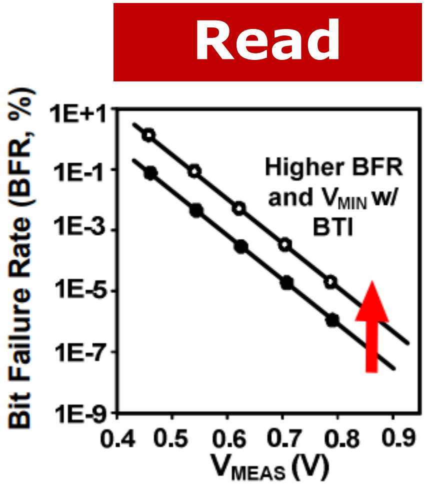


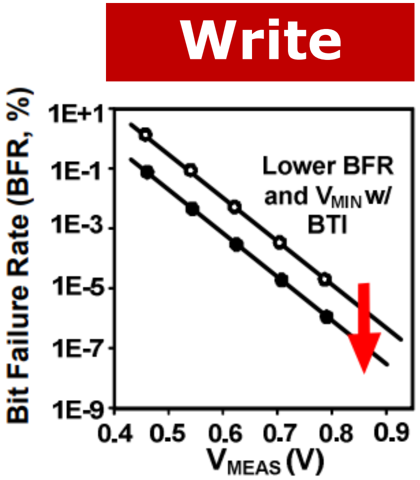
Aging Effects

- Bias Temperature Instability (BTI) effects drive Vmin failure mechanisms
  - SRAM Read Stability degrades with NBTI (PMOS weakening)
  - SRAM Write Margin improves with NBIT (PMOS weakening)
- The effect is statistical in nature, and margining for it requires understanding:
  - Technology characteristics (Intrinsic BTI or other aging effects)
  - Stress conditions (voltage, temperature, etc.)
  - Memory array size



DC stress conditions
NBTI PBTI



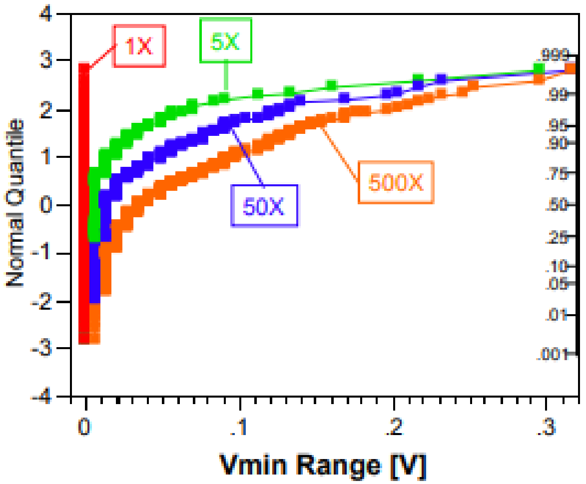


Source: Jain, IEDM 2012

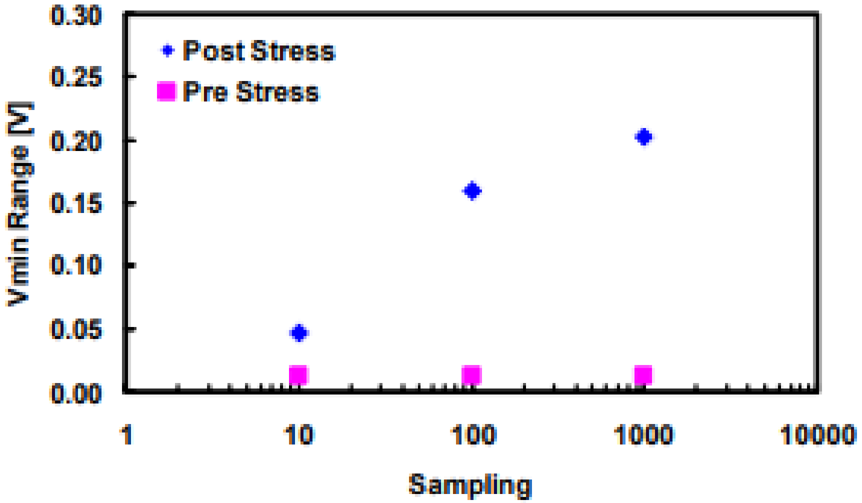
Erratic Bits: Random Telegraph Noise

- Erratic bit phenomena have been observed in SRAM and attributed to trapping/detrapping effects that modify transistor threshold voltage
- Depending upon technology optimization, operating voltage range and usage application, voltage guardbands may be required to avoid in-field errors

Distribution of SRAM Array Vmin with Repeated Measurement



Repeated Single Cell Measurement of Erratic Bit



Source: Agostinelli, IEDM 2005

Memory Repair and ECC

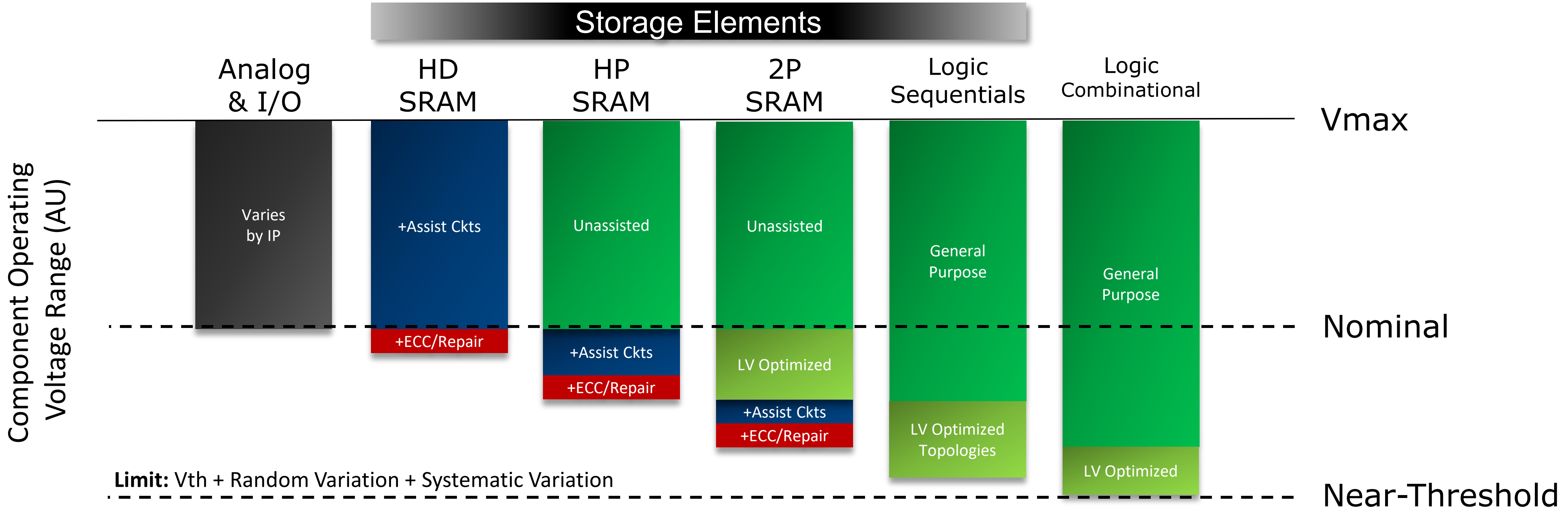
- Repair schemes to replace outlier memory cells with alternate cells are an extremely effective approach to mitigate numerous memory failure mechanisms
  - Manufacturing Defects / Yield
  - Minimum Operating Voltage Limiters
  - Performance Outliers
- Error-correcting codes can take bit error rate (BER) reductions further, by fixing 1-2 errors per data word
- 2-4 order of magnitude reductions in BER possible with minimal overhead

Technique	Maximum BER	Protects tags?	Area Overhead		Total Cache Overhead
			Tags	Data	
Proposed (BB +DCR+LD) §	9.8×10^{-5}	Yes	BB: 15% DCR: 6% LD: 4%	DCR: 0.7%	2.2%
Extreme Redundancy**	2.4×10^{-5}	Yes	SC: 6.4% Flops: 0.5%	SC: 6.4% Flops: 0.5%	6.6%
Line Disable § [3]	1.8×10^{-5}	No	4% #	-	0.2% #
ECC† [4]	1.3×10^{-6}	Yes	7% #	7%	6.7% #
Static Redundancy* [2]	4.4×10^{-7}	Yes	SC: 1% # Fuses: 0.12%	SC: 1% # Fuses: 0.12%	1.1% #
Nominal	1.1×10^{-10}	No	-	-	-

§ Up to 1% disabled **1 column repair/KB and 1 row repair/32KB
†1 repair/128 bit *10 repairs/MB
Estimate (not reported), fuse=100× bitcell, flop=10× bitcell
Note: Data portion is 90% of cache area, tag portion is 6% of cache area

Source: Zimmer, ASSCC 2016

A Roadmap to Low Voltage Memory



- Assisted SRAM enables operation below nominal
- 2P SRAM, with independent optimization for read/write ports, can go further
- Optimized Logic Sequentials can reach near-threshold regime

Conclusion

- Embedded memory usage is increasing due to off-package bandwidth challenges and emerging memory-centric computing models
- Energy efficiency and low voltage operation of memory elements is a critical consideration for high throughput compute systems of the future
- A rough roadmap for memory solutions:
  - SRAM designers have tools to take Vmin below nominal operating voltages
  - Customized, application specific multiport memories can take a further step
  - Logic Sequential arrays can be designed to reach near-threshold operation
- System designers have additional tools to enable Vmin
  - Dual-Rail supply schemes
  - Vmin Repair
  - Error Correcting Code Schemes

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